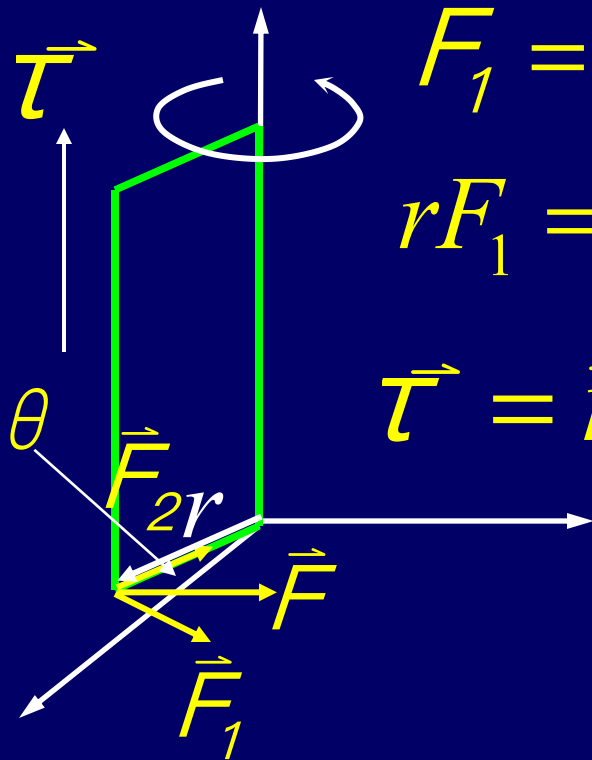


Chapter 9

Rotational Dynamics

Torque

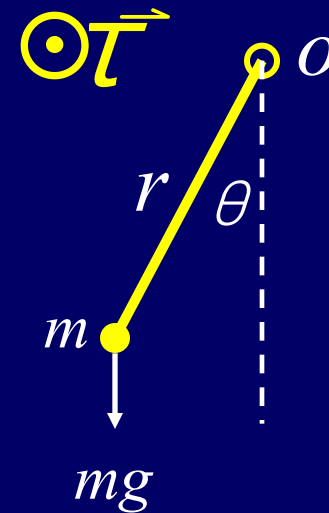
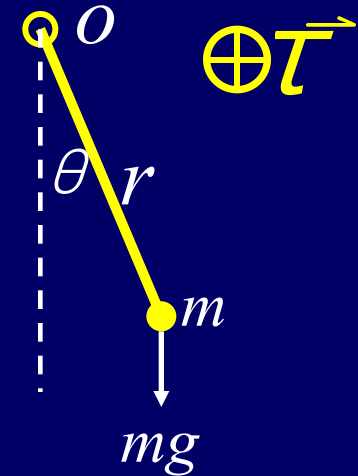


$$F_1 = F \sin \theta$$

$$rF_1 = rF \sin \theta = \tau$$

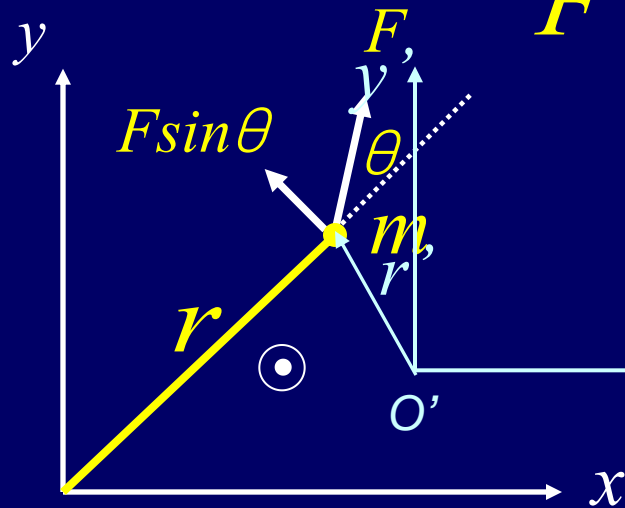
$$\vec{\tau} = \vec{r} \times \vec{F}$$

$\vec{\tau}$ and $\vec{F}$ in different directions



Rotational inertia and Newton's law

A single particle



$$F \sin \theta = m a_T = m a r$$

$$r F \sin \theta = m r^2 \alpha$$

$$\vec{\tau}' = \vec{r}' \times \vec{F} \quad \mathbf{I} \quad I' = m r'^2$$

We define quantity $m r^2$ to be the **rotational inertia** I of the particle:

Rotational motion

$$\vec{\tau}' = I' \vec{a}$$

Linear motion

$$\vec{F} = m \vec{a}$$

$\vec{\tau}$ and I are
dependent on r
 $\vec{F}$ and m are not

*The rotational inertia depends on both the mass of the particle and the perpendicular **distance between the particle and the rotation axis.***

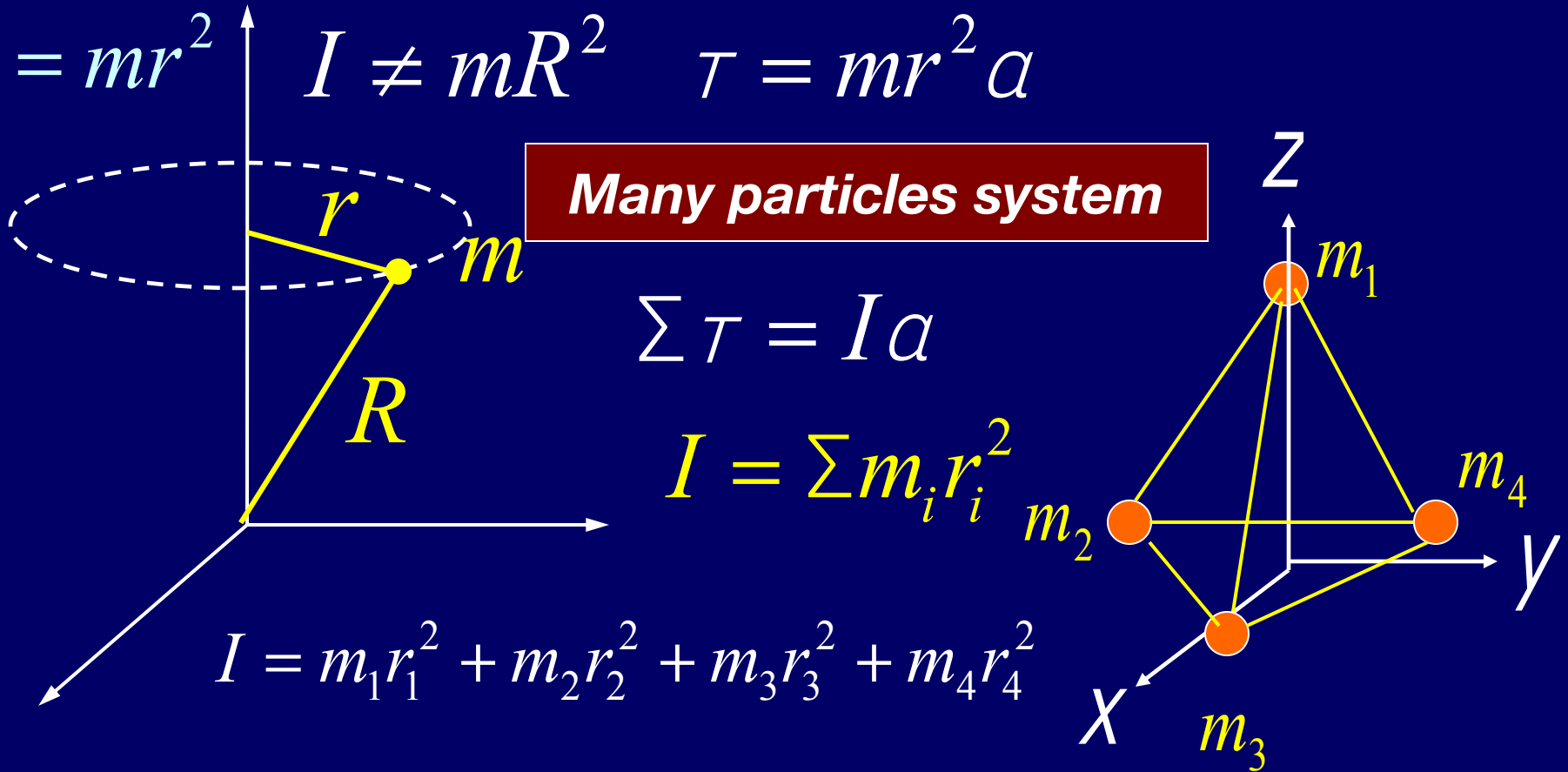
$$I = mr^2 \quad I \neq mR^2 \quad \tau = mr^2 \alpha$$

Many particles system

$$\Sigma \tau = I \alpha$$

$$I = \Sigma m_i r_i^2$$

$$I = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2$$



Rotational inertia of solid bodies

$$I = \sum r_i^2 \delta m_i$$

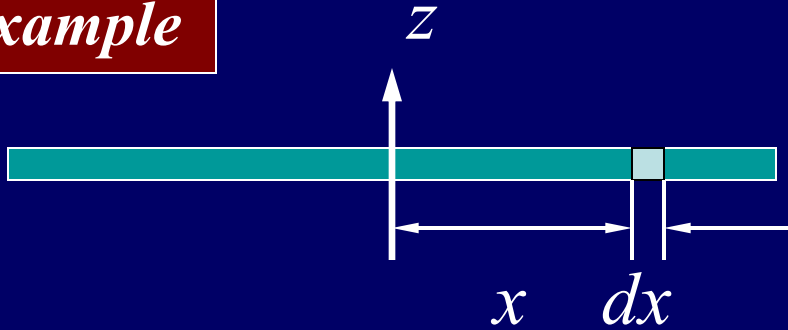
$$I = \lim_{\delta m_n \rightarrow 0} \sum r_n^2 \delta m_n = \int r^2 dm$$

$$dm = \rho dV$$

$$dm = \sigma dS$$

$$dm = \lambda dl$$

Example

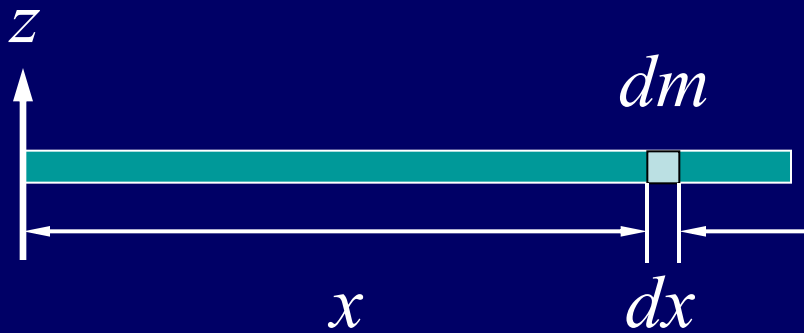


A thin rod with mass M , it is uniform density λ , and length L rotate about z axis through the center of mass.

$$I_c = \int x^2 dm = \int_{-L/2}^{L/2} x^2 \lambda dx = \int_{-L/2}^{L/2} x^2 \frac{M}{L} dx$$

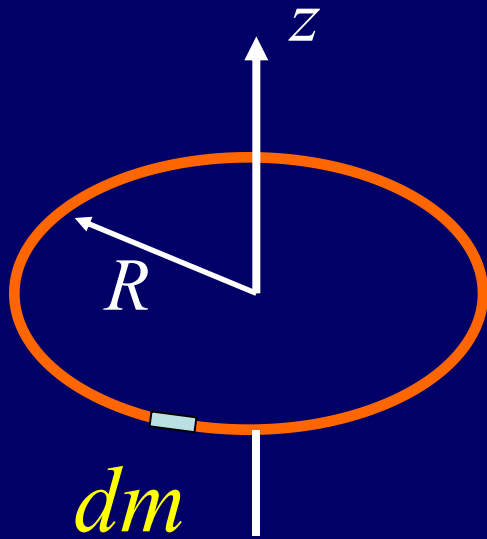
$$= \frac{M}{L} \frac{1}{3} \left[\left(\frac{L}{2} \right)^3 - \left(-\frac{L}{2} \right)^3 \right] = \frac{1}{12} ML^2$$

Example



If the axis through the one end of the rod perpendicular to length is parallel with axis through the center of mass.

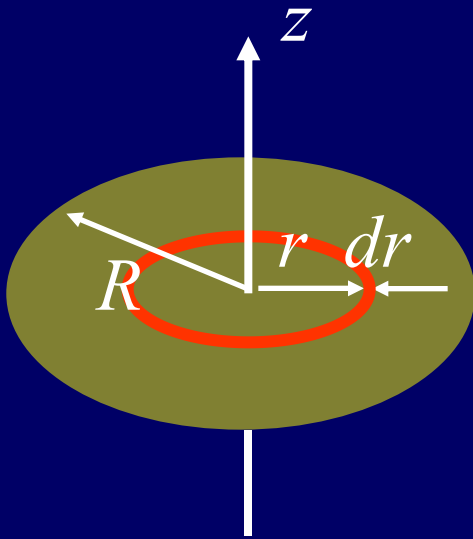
$$\begin{aligned} I &= \int x^2 dm = \int_0^L x^2 \lambda dx = \int_0^L x^2 \frac{M}{L} dx \\ &= \frac{M}{L} \frac{1}{3} L^3 = \frac{1}{3} ML^2 \end{aligned}$$



Example

A ring with radius R , and mass M rotates about z axis perpendicular to the plate.

$$\begin{aligned} I &= \int r^2 dm = \int R^2 dm \\ &= R^2 \int dm = MR^2 \end{aligned}$$

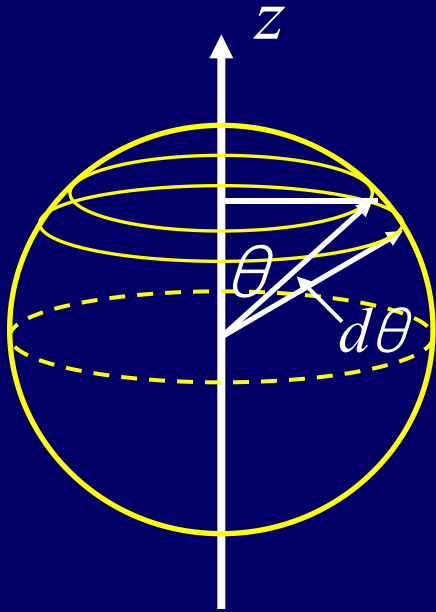


Example

A merry-go-round with radius R and mass M rotates about the z axis perpendicular to the plate.

$$\begin{aligned} I &= \int r^2 dm = \int r^2 \sigma ds = \int_0^R r^2 \frac{M}{\pi R^2} 2\pi r dr \\ &= \frac{2M}{R^2} \frac{R^4}{4} = \frac{1}{2} MR^2 \end{aligned}$$

Example



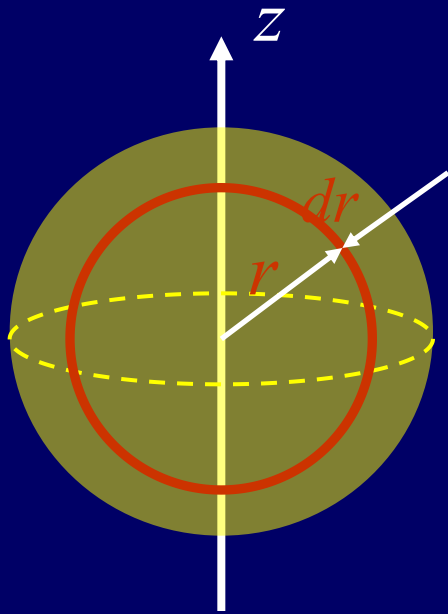
A thin spherical shell with radius R and mass M rotates about the z axis.

$$I = \int r^2 dm = \int r^2 \sigma ds$$

$$= \int_0^\pi (R \sin \theta)^2 \frac{M}{4\pi R^2} 2\pi \cdot R \sin \theta \cdot R d\theta$$

$$= \int_0^\pi (R \sin \theta)^2 \frac{M}{4\pi R^2} 2\pi R \sin \theta R d\theta$$

$$= \frac{2}{3} MR^2$$

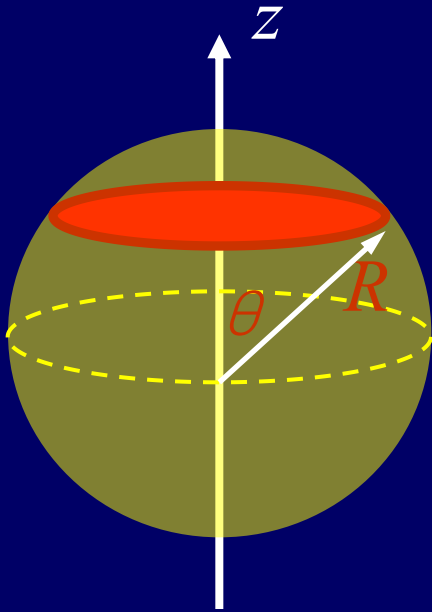


Example

A solid ball with radius R and mass M rotates about the z axis.

$$\begin{aligned} I &= \int \frac{2}{3} r^2 dm = \frac{2}{3} \int r^2 \rho dV \\ &= \frac{2}{3} \int_0^R r^2 \frac{M}{\frac{4}{3} \pi R^3} 4\pi r^2 dr \\ &= \frac{2}{5} MR^2 \end{aligned}$$

Example



A solid ball with radius R and mass M rotates about the z axis.

$$I = \int \frac{1}{2} r^2 dm = \frac{1}{2} \int r^2 \rho dV$$

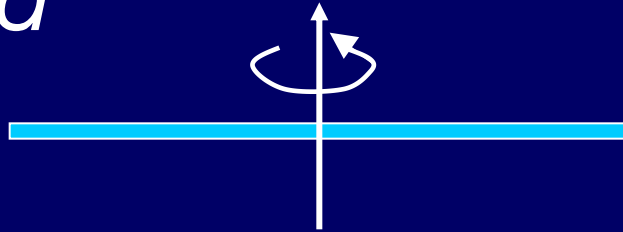
$$= -\frac{1}{2} \int_0^\pi (R \sin \theta)^2 \frac{M}{(4/3)\pi R^3} \pi (R \sin \theta)^2 R \sin \theta d\theta$$

$$z = R \cos \theta$$

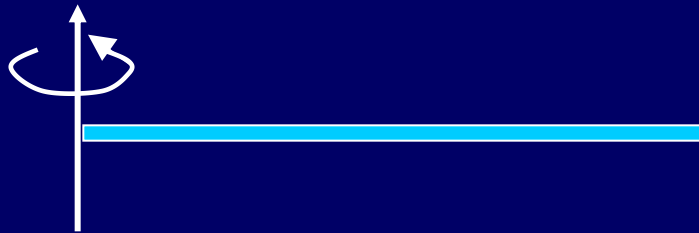
$$dz = -R \sin \theta d\theta$$

$$= \frac{2}{5} MR^2$$

rod

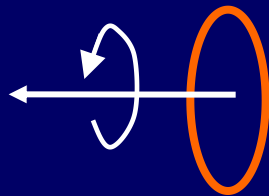


$$I = \frac{1}{12} M L^2$$

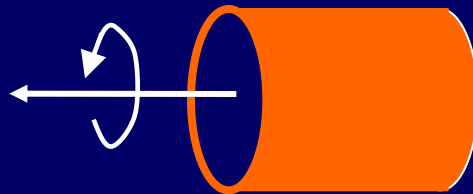


$$I = \frac{1}{3} M L^2$$

ring

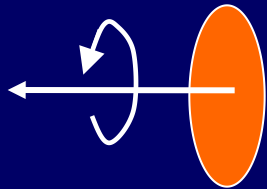


*cylinder
shell*

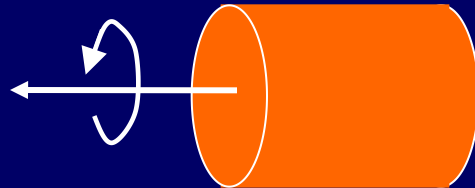


$$\left. \begin{array}{l} \text{ring} \\ \text{cylinder shell} \end{array} \right\} I = M R^2$$

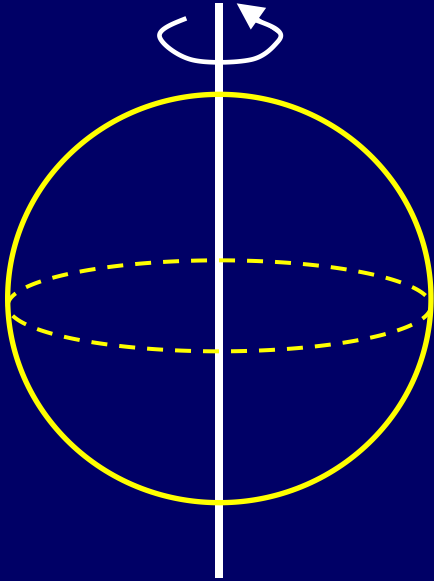
disk



cylinder



$$\left. \begin{array}{l} \text{disk} \\ \text{cylinder} \end{array} \right\} I = \frac{1}{2}MR^2$$



thin spherical shell

$$I = \frac{2}{3}MR^2$$

solid sphere

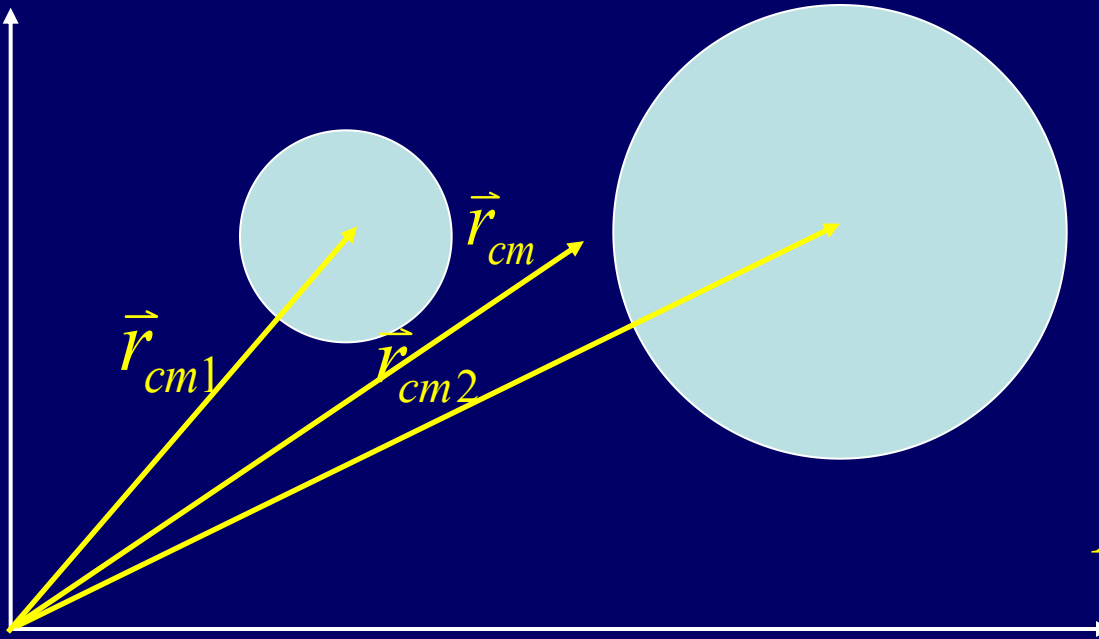
$$I = \frac{2}{5}MR^2$$

Mass center

$$\vec{r}_{cm} = \frac{M_1 \vec{r}_{cm1} + M_2 \vec{r}_{cm2}}{M}$$

Rotational inertia

~~$$I = M_1 r_{cm1}^2 + M_2 r_{cm2}^2$$~~



$$I = \sum r_i^2 m_i = (r_1^2 m_1 + \dots + r_m^2 m_m) + (r_{m+1}^2 m_{m+1} + \dots + r_n^2 m_n)$$

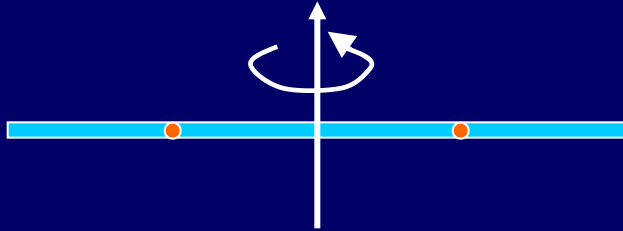
$$= I_1 + I_2$$

~~$$M_1 r_{cm1}^2 + M_2 r_{cm2}^2$$~~

$$= M_1 \left(\frac{m_1 r_1 + m_2 r_2 + \dots + m_m r_m}{M_1} \right)^2 + M_2 \left(\frac{m_{m+1} r_{m+1} + m_{m+2} r_{m+2} + \dots + m_n r_n}{M_2} \right)^2$$

rod

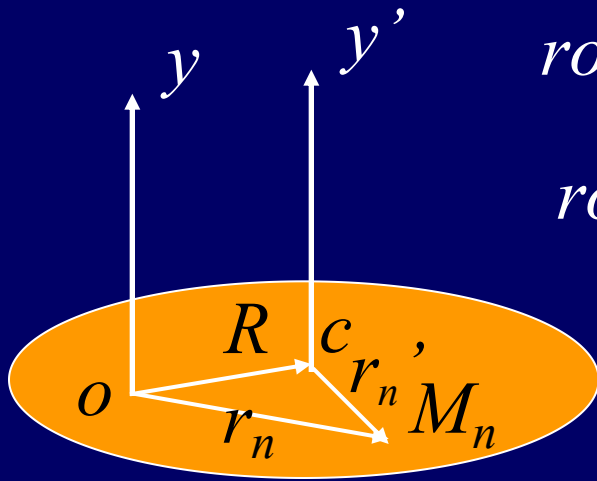
$$I = 2 \times \left[\frac{1}{3} \frac{M}{2} \left(\frac{L}{2} \right)^2 \right]$$



$$I = \frac{1}{12} ML^2$$

$$I = \frac{M}{2} \left(\frac{L}{4} \right)^2 + \frac{M}{2} \left(\frac{L}{4} \right)^2 = \cancel{\frac{1}{16}} ML^2$$

The parallel-axis theorem



rotate about y' axis $I_{cm} = \frac{1}{2}MR_0^2$

rotate about y axis $I = ?$ $I = \sum r_n^2 m_n$

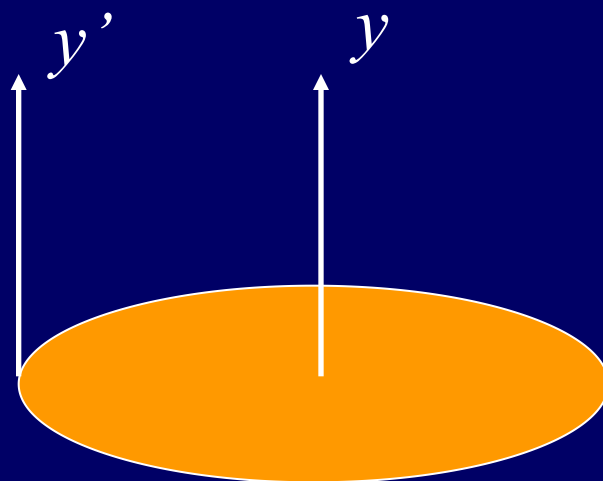
$$I = \sum (\vec{R} + \vec{r}'_n)^2 m_n$$

$$I = \sum R^2 m_n + \sum r_n'^2 m_n + \sum 2\vec{R} \cdot \vec{r}'_n m_n$$

$$\sum R^2 m_n = MR^2 \quad \sum r_n'^2 m_n = I_{cm}$$

$$\sum 2m_n \vec{R} \cdot \vec{r}'_n = 2\vec{R} \cdot \sum m_n \vec{r}'_n = 2\vec{R} \cdot M\vec{r}'_{cm} = 0$$

$$I = MR^2 + I_{cm}$$

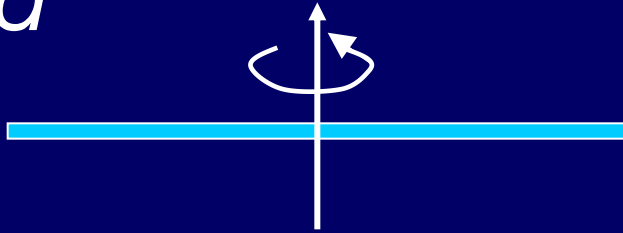


$$I = \frac{1}{2}MR^2$$

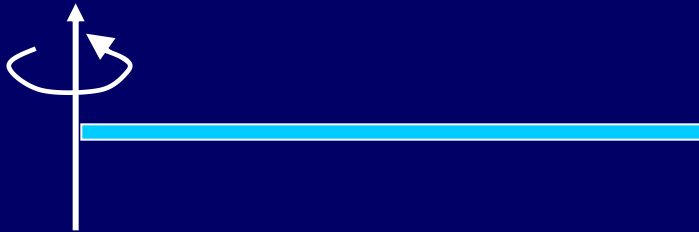
$$I' = MR^2 + \frac{1}{2}MR^2$$

$$= \frac{3}{2}MR^2$$

rod



$$I_c = \frac{1}{12} ML^2$$

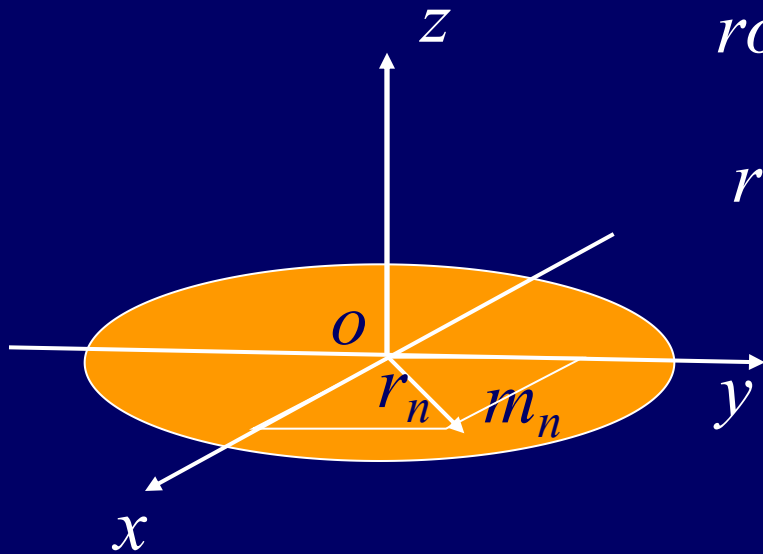


$$I = I_c + M\left(\frac{1}{2}L\right)^2$$

$$= \frac{1}{12} ML^2 + \frac{1}{4} ML^2$$

$$= \frac{1}{3} ML^2$$

The perpendicular-axis theorem



rotate about z axis $I_z = \frac{1}{2}MR^2$
rotate about y axis $I = ?$

$$I_z = \sum (x_n^2 + y_n^2) m_n$$

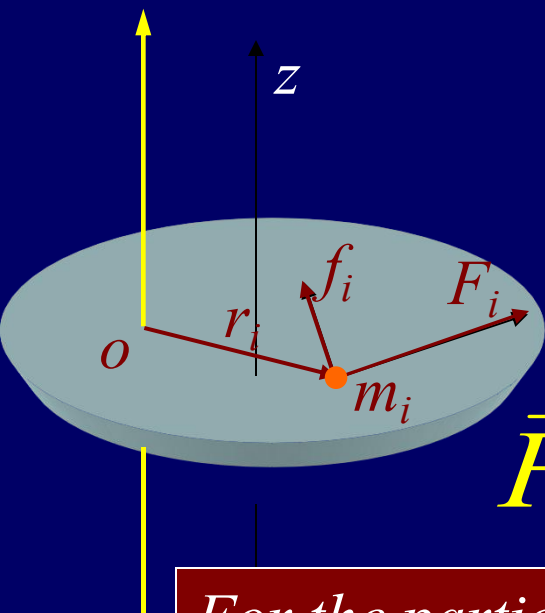
$$I = \sum x_n^2 m_n + \sum y_n^2 m_n$$

$$I_x + I_y = I_z = \frac{1}{2}MR^2$$

$$I_x = I_y = \frac{1}{2}I_z = \frac{1}{4}MR^2$$

$$\boxed{I_z = I_y + I_x}$$

A rigid system with many particles



perpendicular distance is r_i ,

external force F_i

internal forces f_i .

$$\vec{F}_i + \vec{f}_i = m_i \vec{a}_i \quad \Sigma \vec{F}_{ext} = M \vec{a}_{cm}$$

For the particle i

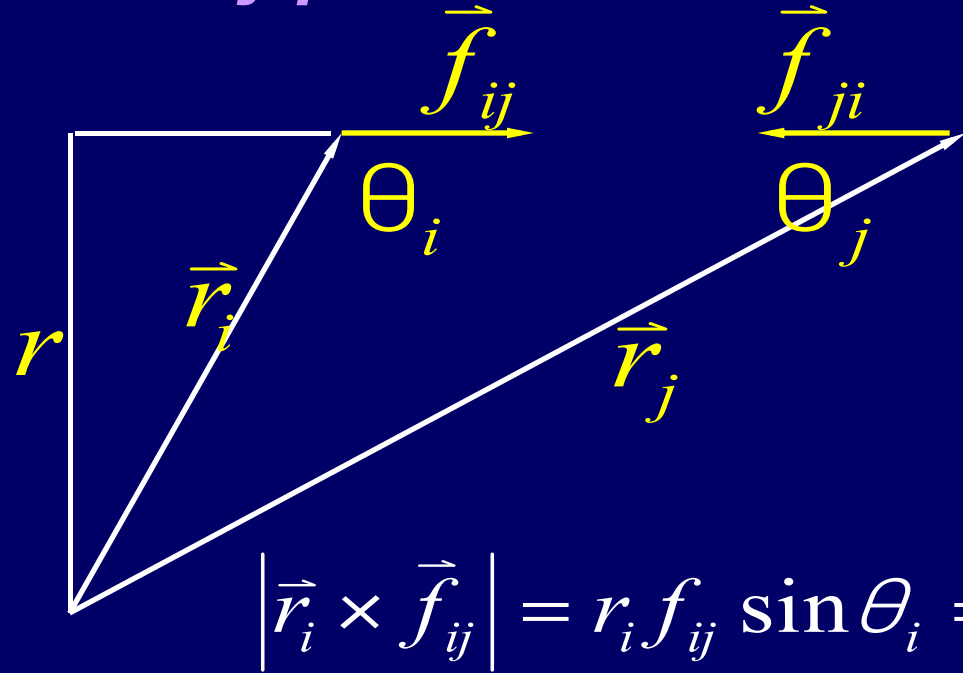
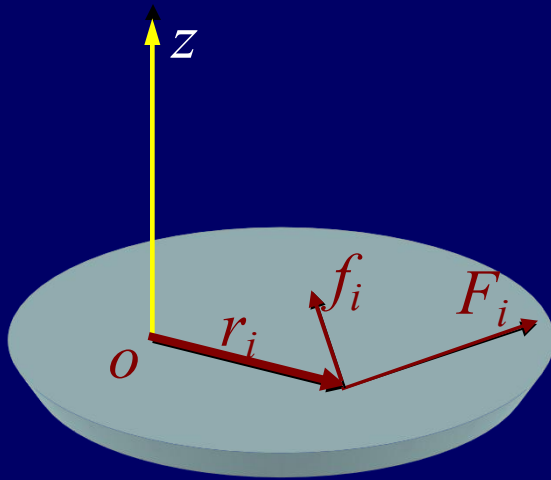
$$\vec{r}_i \times \vec{F}_i + \vec{r}_i \times \vec{f}_i = m_i \vec{r}_i \times \vec{a}_i$$

$$= m_i \vec{r}_i \times (\vec{a}_{Ti} + \vec{a}_{Ri}) = m_i \vec{r}_i \times \vec{a}_{Ti} = m_i \vec{r}_i \times (\vec{a} \times \vec{r}_i)$$

For all particles

$$\Sigma \vec{r}_i \times \vec{F}_i + \Sigma \vec{r}_i \times \vec{f}_i = \Sigma m_i \vec{r}_i \times (\vec{a} \times \vec{r}_i)$$

A rigid system with many particles



$$|\vec{r}_i \times \vec{f}_{ij}| = r_i f_{ij} \sin \theta_i = r f_{ij}$$

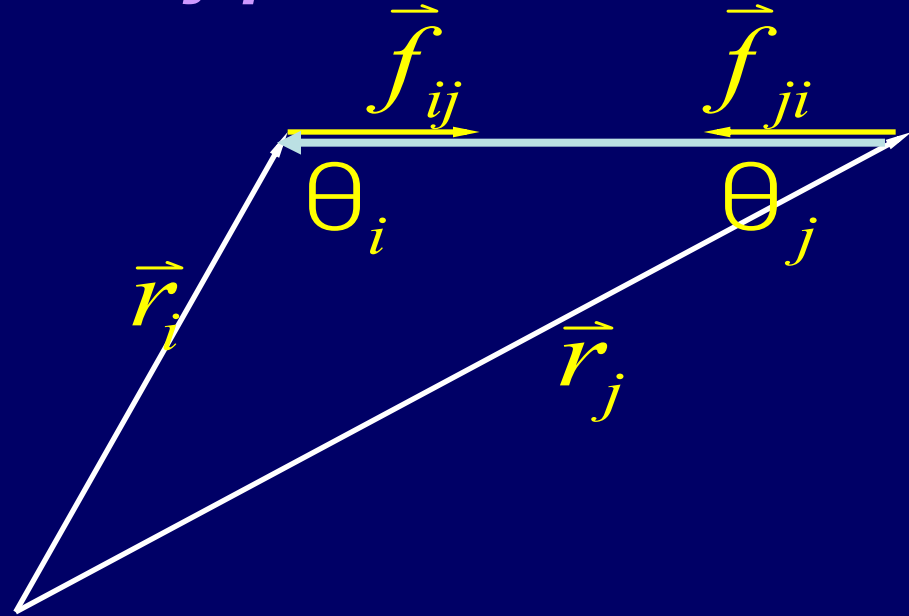
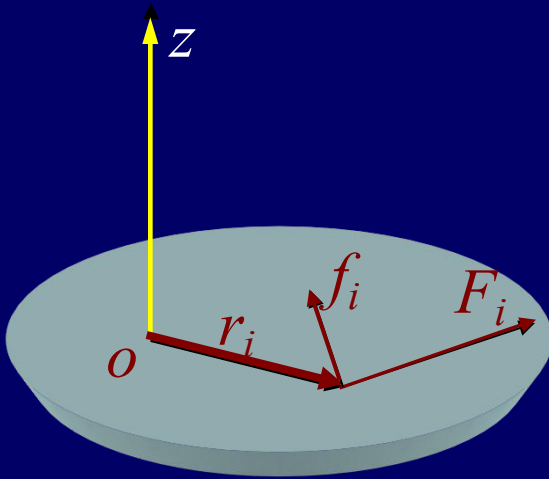
$$\vec{r}_i \times \vec{f}_{ij} = -\vec{r}_j \times \vec{f}_{ji}$$

$$|\vec{r}_j \times \vec{f}_{ji}| = r_j f_{ji} \sin \theta_j = r f_{ji}$$

For all particles

$$\Sigma \vec{r}_i \times \vec{F}_i + \boxed{0} = \Sigma m_i \vec{r}_i \times (\vec{a} \times \vec{r}_i)$$

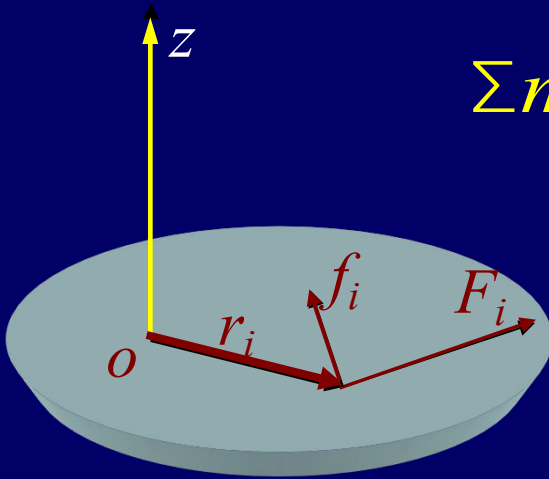
A rigid system with many particles



$$\vec{r}_i \times \vec{f}_{ij} + \vec{r}_j \times \vec{f}_{ji} = (\vec{r}_i - \vec{r}_j) \times \vec{f}_{ij} = 0$$

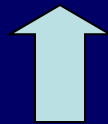
For all particles $\Sigma \vec{r}_i \times \vec{F}_{i\tau} + \boxed{0} = \Sigma m_i \vec{r}_i \times (\vec{a} \times \vec{r}_i)$

A rigid system with many particles



$$\begin{aligned}\sum m_i \vec{r}_i \times (\vec{a} \times \vec{r}_i) &= \sum m_i [(\vec{r}_i \cdot \vec{r}_i) \vec{a} - (\vec{r}_i \cdot \vec{a}) \vec{r}_i] \\ &= \sum m_i r_i^2 \vec{a} \\ &= I \vec{a}\end{aligned}$$

$$\sum \vec{\tau}_{ext} = I \vec{a} \quad \sum \vec{F}_{ext} = M \vec{a}_{cm}$$



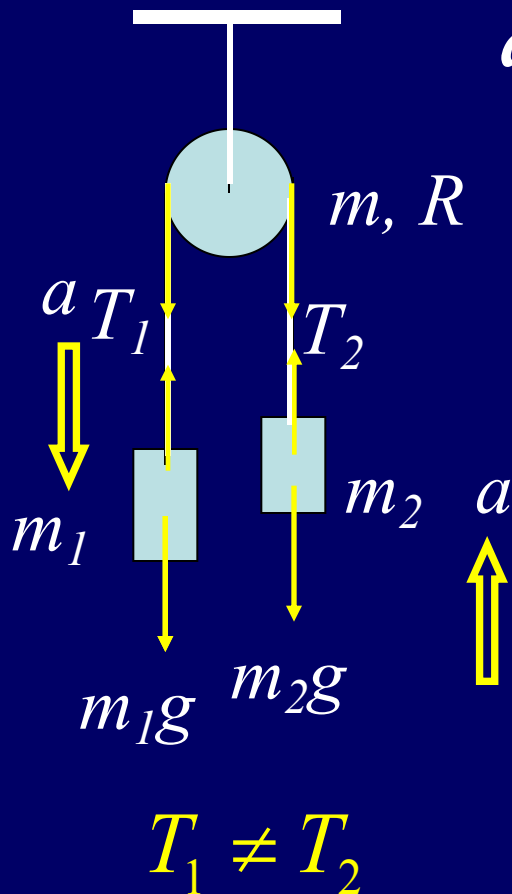
For all particles

$$\left. \sum \vec{r}_i \times \vec{F}_{i\tau} + \right| \quad \quad \quad \mathbf{0} \quad = \sum m_i \vec{r}_i \times (\vec{a} \times \vec{r}_i)$$

Applications of Newton's law for rotation

Example

Find the acceleration of the blocks, the tension in the cord and the angular acceleration of the disk.



if $m_1 > m_2$

$$T_1 = \frac{2m_2 + m/2}{m_1 + m_2 + m/2} m_1 g$$

$$m_1 g - T_1 = m_1 a$$

$$T_2 - m_2 g = m_2 a$$

$$T_2 = \frac{2m_1 + m/2}{m_1 + m_2 + m/2} m_2 g$$

$$a = \frac{m_1 - m_2}{m_1 + m_2 + m/2} g$$

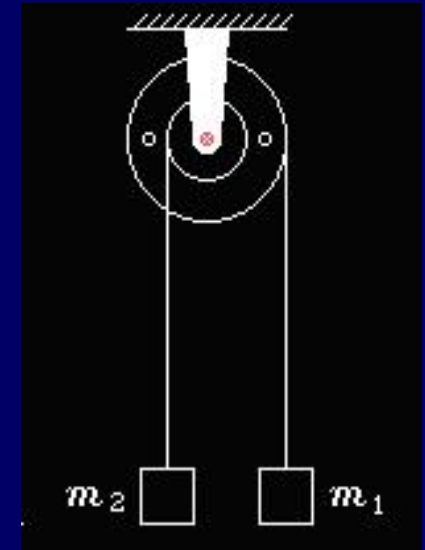
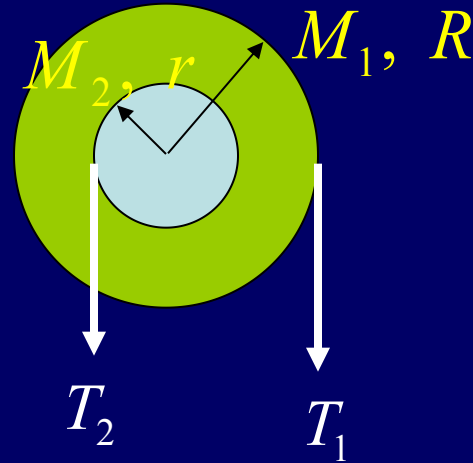
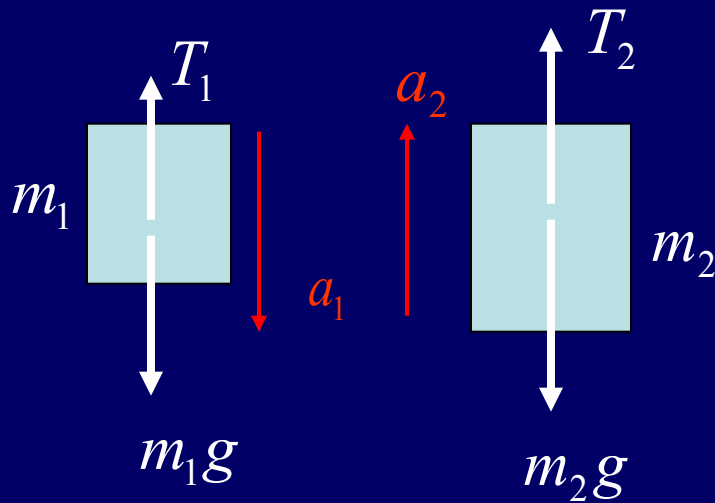
$$\alpha = \frac{a}{R} = \frac{m_1 - m_2}{m_1 + m_2 + m/2} \frac{g}{R}$$

Example

$$M_1 = 6\text{kg}, R = 0.1\text{m}, M_2 = 4\text{kg}, r = 0.05\text{m}$$

$$m_1 = m_2 = 2\text{kg} \quad a_1 = ? \quad a_2 = ?$$

$$T_1 = ? \quad T_2 = ?$$



$$m_1g - T_1 = m_1a_1 \quad \text{--- (1)}$$

$$T_2 - m_2g = m_2a_2 \quad \text{--- (2)}$$

$$T_1R - T_2r = I\alpha \quad \text{--- (3)}$$

$$I = \frac{1}{2}M_1R^2 + \frac{1}{2}M_2r^2$$

$$a_1 = R\alpha$$

$$a_2 = r\alpha$$

$$a_1 = 1.63 \text{ m} \cdot \text{s}^{-2}, \quad a_2 = 0.82 \text{ m} \cdot \text{s}^{-2}$$

$$T_1 = 16.3 \text{ N}, \quad T_2 = 21.2 \text{ N}$$

One particle

$$\vec{F} = m\vec{a} \quad \vec{\tau} = I\vec{\alpha}$$

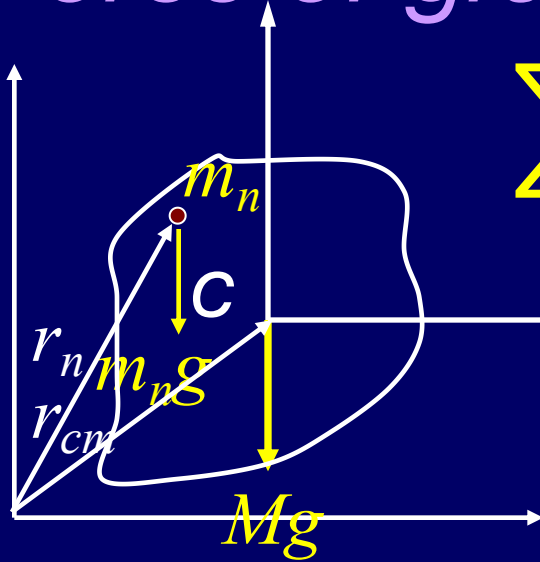
$$\vec{\tau} = \vec{r} \times \vec{F} \quad I = mr^2$$

Many particles

$$\Sigma \vec{F}_{ext} = M\vec{a}_{cm} \quad \Sigma \vec{\tau}_{ext} = I\vec{\alpha}$$

$$\Sigma \vec{\tau}_{ext} = \Sigma \vec{r}_i \times \vec{F}_{ext,i} \quad I = \Sigma m_i r_i^2$$

Force of gravity



$$\sum \vec{F} = \sum m_n \vec{g} = \vec{g} \sum m_n = M\vec{g}$$

$$\begin{aligned} \sum \vec{\tau}_n &= \sum (\vec{r}_n \times m_n \vec{g}) = \sum (m_n \vec{r}_n \times \vec{g}) \\ &= \left(\sum m_n \vec{r}_n \right) \times \vec{g} = M\vec{r}_{cm} \times \vec{g} \end{aligned}$$

$$\vec{\tau}_n = \vec{r}_n \times m_n \vec{g} = \vec{r}_{cm} \times M\vec{g}$$

cm frame

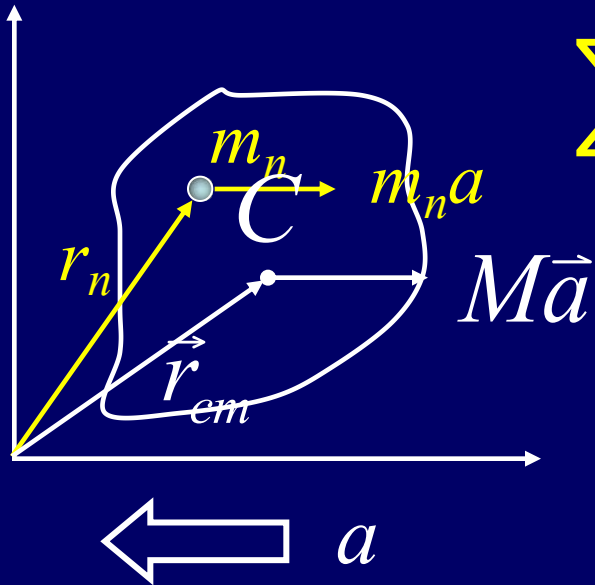
$$\vec{r}'_{cm} = 0$$

$$\sum \vec{\tau}'_n = 0$$

A useful corollary: the torque due to gravity about the center of mass of a body is zero.

Torque from pseudoforce

Same as torque due to gravity

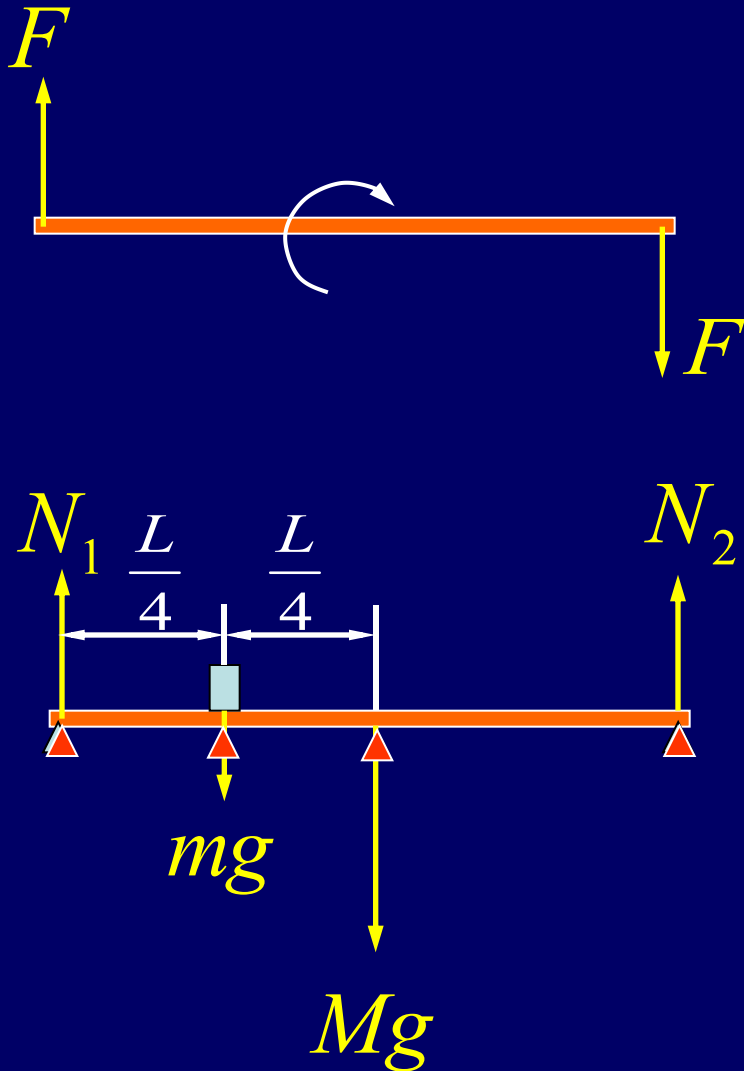


$$\begin{aligned}\sum \vec{\tau}_n &= \sum (\vec{r}_n \times m_n \vec{a}) = \sum (m_n \vec{r}_n \times \vec{a}) \\ &= (\sum m_n \vec{r}_n) \times \vec{a} = M \vec{r}_{cm} \times \vec{a} \\ &= \vec{r}_{cm} \times M \vec{a}\end{aligned}$$

$$\vec{\tau}_n = \vec{r}_n \times m_n \vec{a}$$

*The center of
pseudoforce coincides
with the center of mass*

Condition of Equilibrium



$$\begin{cases} \vec{a}_{mc} = 0 \\ \vec{a} = 0 \end{cases} \quad \begin{cases} \Sigma \vec{F}_{ext} = 0 \\ \Sigma \vec{\tau}_{ext} = 0 \end{cases}$$

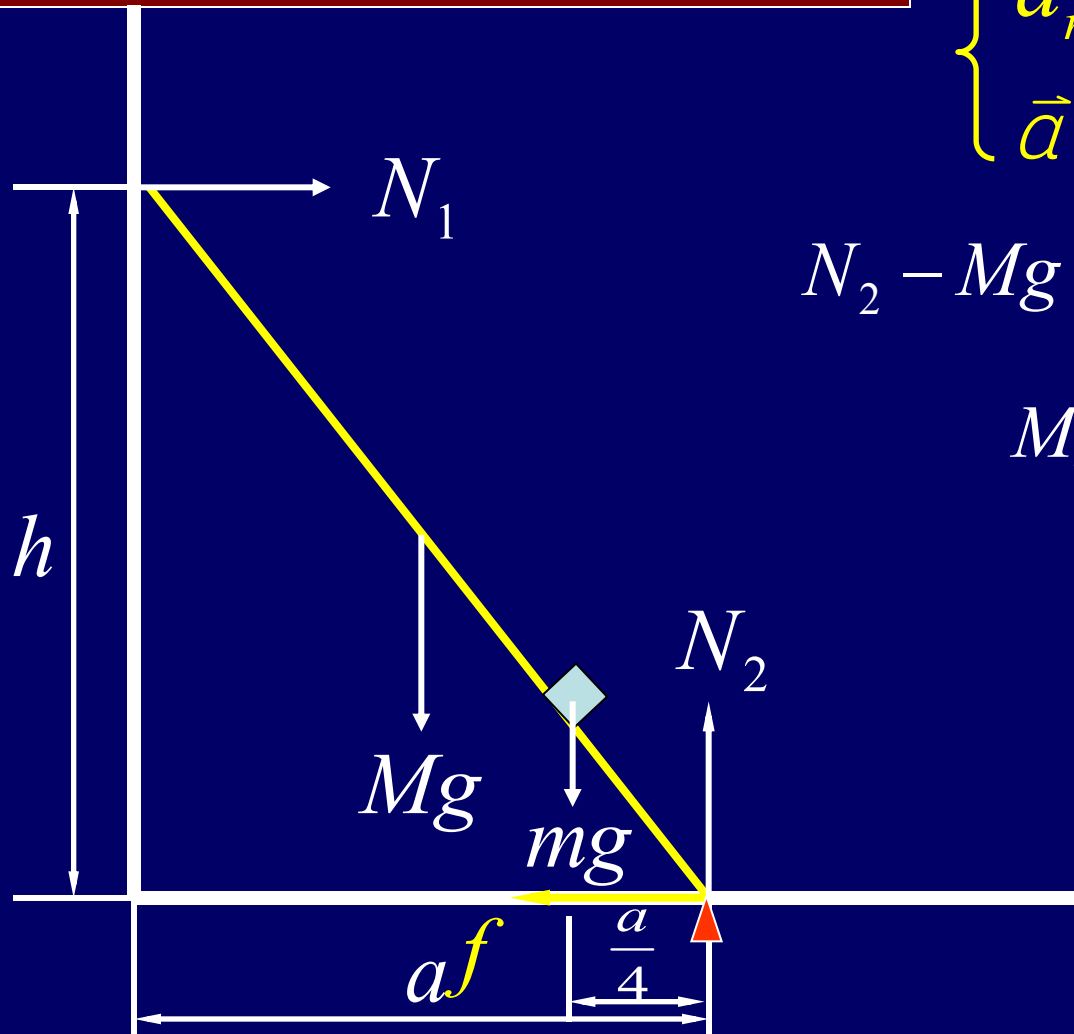
$$N_1 + N_2 - mg - Mg = 0$$

$$N_2 L - mg \frac{L}{4} - Mg \frac{L}{2} = 0$$

$$N_2 = \left(\frac{m}{4} + \frac{M}{2} \right) g$$

$$N_1 = \left(\frac{3m}{4} + \frac{M}{2} \right) g$$

Condition of Equilibrium

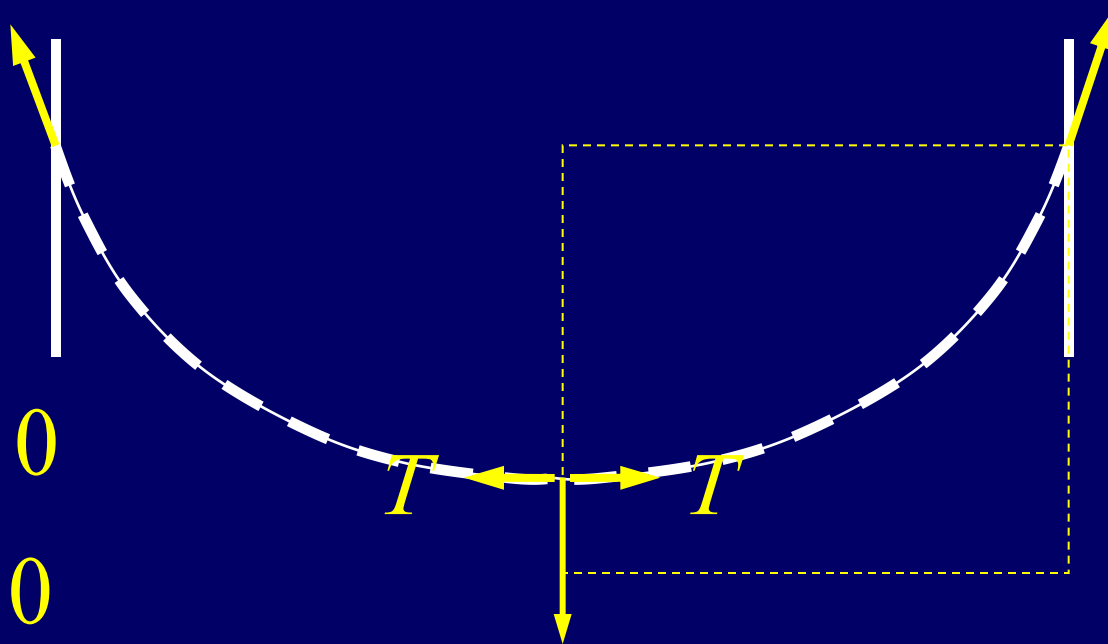


$$\begin{cases} \vec{a}_{mc} = 0 \\ \vec{a} = 0 \end{cases} \quad \begin{cases} \Sigma \vec{F}_{ext} = 0 \\ \Sigma \vec{\tau}_{ext} = 0 \end{cases}$$

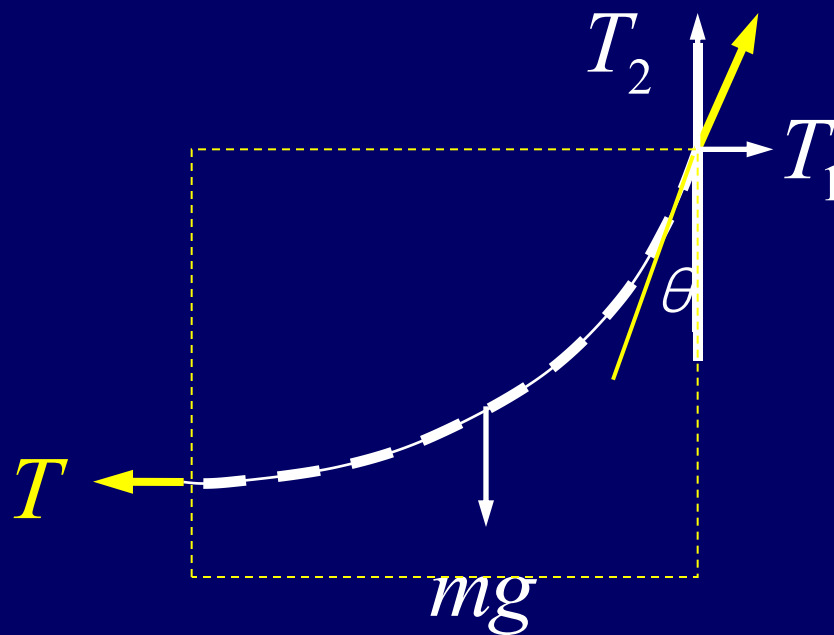
$$N_2 - Mg - mg = 0 \quad N_1 - f = 0$$

$$Mg \frac{a}{2} + mg \frac{a}{4} - N_1 h = 0$$

$$\begin{cases} \Sigma \vec{F}_{ext} = 0 \\ \Sigma \vec{\tau}_{ext} = 0 \end{cases}$$



$$T = ?$$



$$\begin{cases} T_1 - T = 0 \\ T_2 - mg = 0 \end{cases}$$

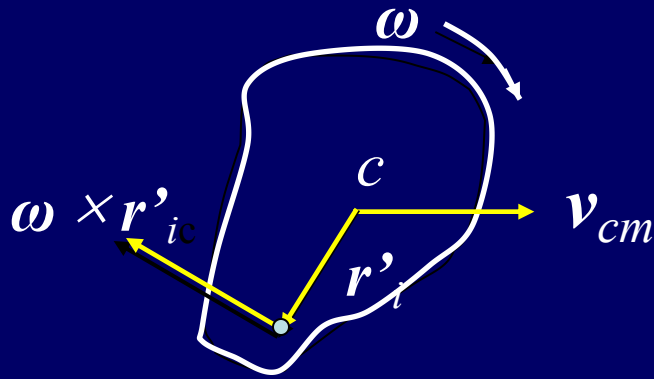
$$T_1 = T_2 \operatorname{tg} \theta$$

$$T = mg \operatorname{tg} \theta$$

Combined rotational and translational motion

The **center of mass** of the object moves translational motion.

The motion of **any other point** of object is a complicated combination of the rotational and translational motion.

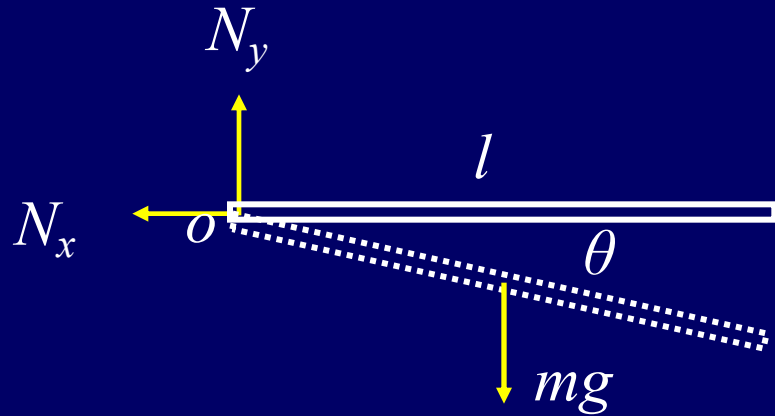


$$\begin{cases} \Sigma \mathbf{F}_{ext} = m \mathbf{a}_{cm} \\ \Sigma \boldsymbol{\tau}_{ext} = I \boldsymbol{\alpha} \end{cases}$$

$$\vec{v}_i = \vec{v}_{cm} + \vec{\omega} \times \vec{r}'_i$$

Example

A rod, mass m and length l , can rotate about point o . The rod rotates from the rest. When it is θ to horizontal, find (1) α ; (2) ω ; (3) a_{cm} ; (4) $N_x N_y$.



$$\int_0^{\omega} \omega d\omega = \int_0^{\theta} \frac{3g}{2l} \cos\theta d\theta$$

$$\omega = \sqrt{\frac{3g \sin\theta}{l}}$$

$$(1) \quad mg \frac{l}{2} \cos\theta = \frac{1}{3} ml^2 \alpha$$

$$\alpha = \frac{3g}{2l} \cos\theta$$

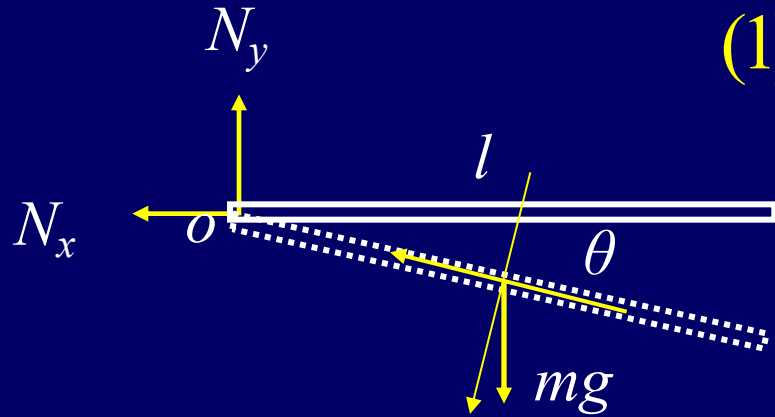
$$(2) \quad \alpha = \frac{d\omega}{dt} = \frac{3g}{2l} \cos\theta$$

$$\frac{d\omega}{d\theta} \frac{d\theta}{dt} = \frac{3g}{2l} \cos\theta$$

$$\omega d\omega = \frac{3g}{2l} \cos\theta d\theta$$

Example

A rod, mass m and length l , can rotate about point o . The rod rotates from the rest. When it is θ to horizontal, find (1) α ; (2) ω ; (3) a_{cm} ; (4) N_x, N_y .



$$(1) \alpha = \frac{3g}{2l} \cos \theta \quad (2) \omega = \sqrt{\frac{3g \sin \theta}{l}}$$

$$(3) a_{cT} = \alpha \frac{l}{2} = \frac{3g}{4} \cos \theta$$

$$a_{cR} = \omega^2 \frac{l}{2} = \frac{3g}{2} \sin \theta$$

$$(4) N_x \sin \theta - N_y \cos \theta + mg \cos \theta = ma_{cT}$$

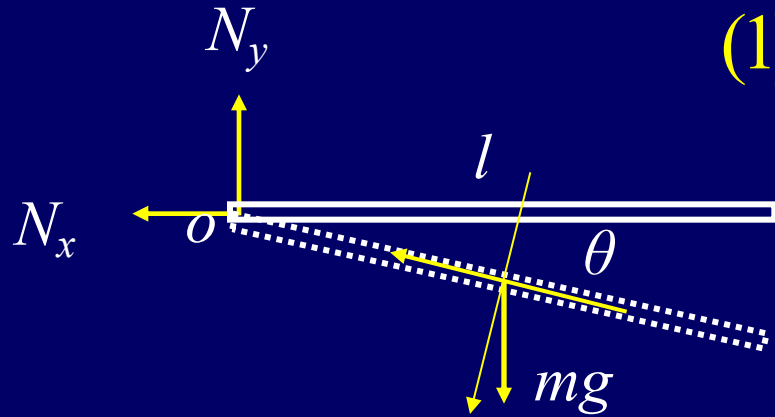
$$N_x \cos \theta + N_y \sin \theta - mg \sin \theta = ma_{cR}$$

$$N_x = \frac{9}{4} mg \sin \theta \cos \theta$$

$$N_y = mg \left(1 + \frac{3}{2} \sin^2 \theta - \frac{3}{4} \cos^2 \theta \right)$$

Example

A rod, mass m and length l , can rotate about point o . The rod rotates from the rest. When it is θ to horizontal, find (1) α ; (2) ω ; (3) a_{cm} ; (4) N_x, N_y .



$$(1) \alpha = \frac{3g}{2l}$$

$$(2) \omega = 0$$

$$(3) a_{cT} = \alpha \frac{l}{2} = \frac{3g}{4}$$

$$a_{cR} = \omega^2 \frac{l}{2} = 0$$

$$(4) N_x \sin\theta - N_y \cos\theta + mg \cos\theta = ma_{cT}$$

$$N_x \cos\theta + N_y \sin\theta - mg \sin\theta = ma_{cR}$$

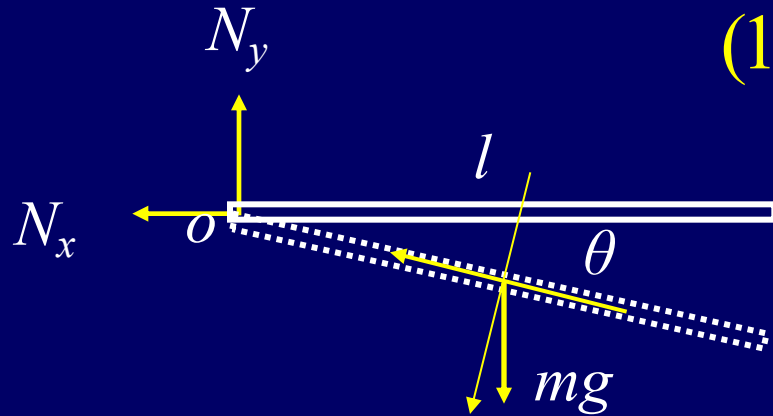
$$N_x = 0$$

$$N_y = \frac{mg}{4}$$

When $\theta=0^\circ$

Example

A rod, mass m and length l , can rotate about point o . The rod rotates from the rest. When it is θ to horizontal, find (1) α ; (2) ω ; (3) a_{cm} ; (4) N_x, N_y .



$$(1) \alpha = 0$$

$$(2) \omega = \sqrt{\frac{3g}{l}}$$

$$(3) a_{cT} = \alpha \frac{l}{2} = 0$$

$$a_{cR} = \omega^2 \frac{l}{2} = \frac{3g}{2}$$

$$(4) N_x \sin\theta - N_y \cos\theta + mg \cos\theta = ma_{cT}$$

$$N_x \cos\theta + N_y \sin\theta - mg \sin\theta = ma_{cR}$$

$$N_x = 0$$

$$N_y = \frac{5}{2} mg$$

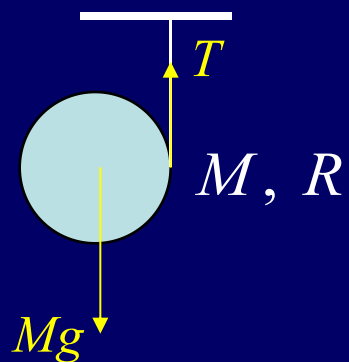
When $\theta=0^\circ$

When $\theta=90^\circ$

plane parallel motion

Now we restrict our discussion of a combined motion that satisfies two conditions: (1) the axis of rotation passes through the center of mass, and (2) the axis always has the same direction in space. We call this motion as the **plane parallel motion**.





$a, a, T?$

$$Mg - T = Ma \quad a = R\alpha$$

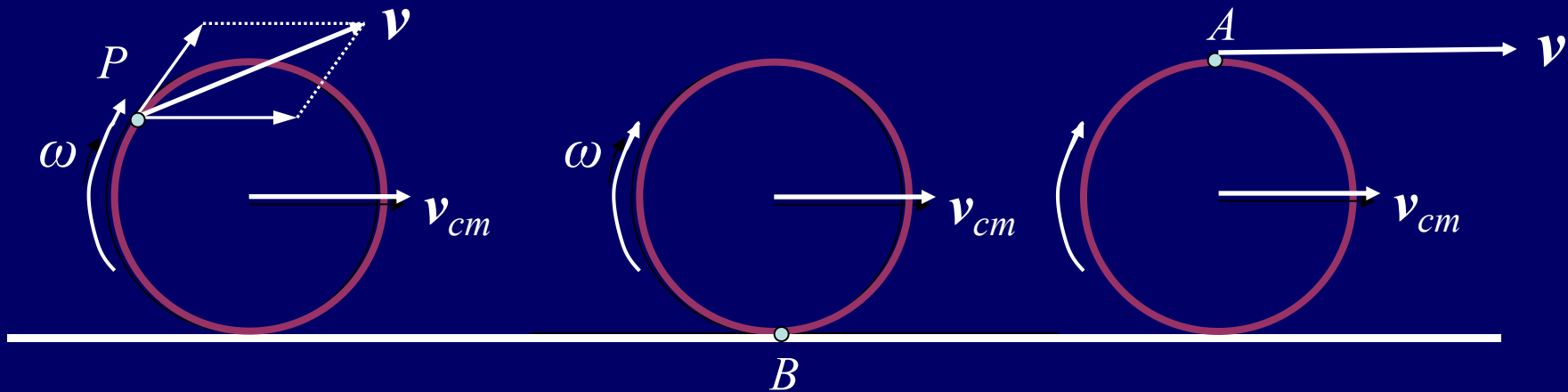
$$TR = \frac{1}{2}MR^2\alpha \quad T = \frac{1}{2}Ma$$

$$Mg = \frac{3}{2}Ma$$

$$a = \frac{2}{3}g \quad a = \frac{2}{3}\frac{g}{R}$$

$$T = \frac{1}{3}Mg$$

Rolling without slipping



$$\vec{v}_i = \vec{v}_{cm} + \vec{\omega} \times \vec{r}_i' \quad v_B = v_{cm} - \omega R = 0 \quad v_A = v_{cm} + \omega R = 2v_{cm}$$

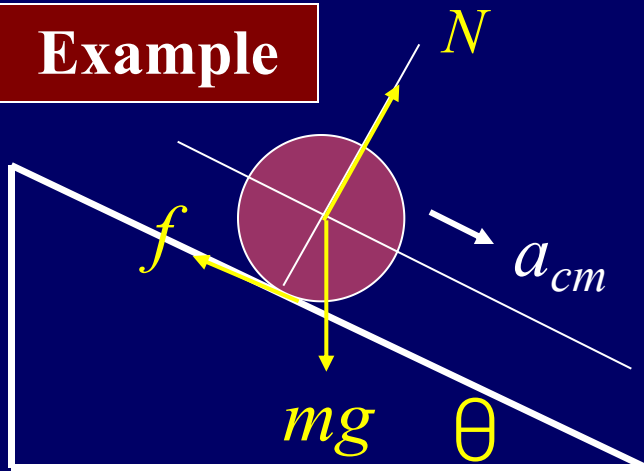
There is not relative motion at the point of contact.

$$v_{cm} = \omega R = 2\pi R / T$$

the condition of rolling without slipping.

$$v_{cm} = \omega R \quad a_{cm} = aR$$

Example



A solid cylinder of mass m and radius R rolls without slipping down an inclined plane with angle θ . Find the a_{cm} and f .

motion of the center of mass:

$$N - mg \cos \theta = 0 \quad mg \sin \theta - f = ma_{cm}$$

For the rotational motion of the cylinder about the axis

$$f = \frac{1}{2} m R \alpha = \frac{1}{2} m a_{cm} \quad a_{cm} = \frac{2}{3} g \sin \theta \quad \alpha = \frac{2g}{3R} \sin \theta$$

Rolling without slipping.

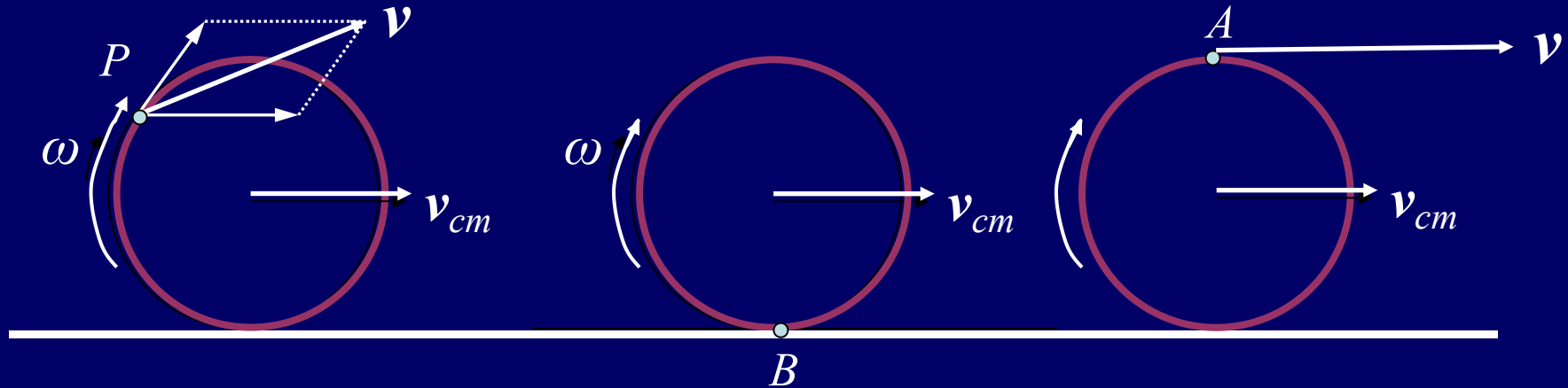
$$(a_{cm} = \alpha R) \quad f = \frac{1}{3} mg \sin \theta \leq \mu_s mg \cos \theta$$

If not, the cylinder may be slipping.

$$\mu_s \geq \frac{1}{3} \tan \theta$$

$\mu_s = 0$, it is complete slipping (no rotate)

Rolling without slipping



$$\vec{v}_i = \vec{v}_{cm} + \vec{\omega} \times \vec{r}_i' \quad v_B = v_{cm} - \omega R = 0 \quad v_A = v_{cm} + \omega R = 2v_{cm}$$

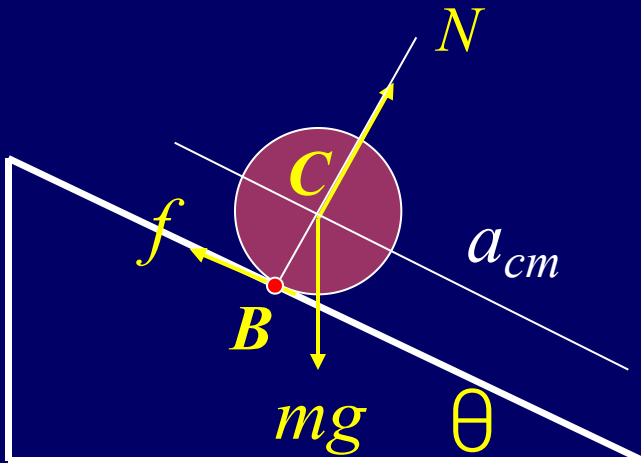
There is not relative motion at the point of contact.

the condition of rolling without slipping.

About **B** point
Instantaneous axis

Example

A solid cylinder of mass m and radius R rolls **without slipping** down an inclined plane with angle θ . Find the a_{cm} and f .



$$\tau = I\alpha$$

$$Rmg\sin\theta = \left(\frac{3}{2}mR^2\right)\alpha$$

$$\alpha = \frac{2g}{3R} \sin\theta$$

About C or B ?

$$a_{cm} = \alpha R = \frac{2g}{3} \sin\theta$$

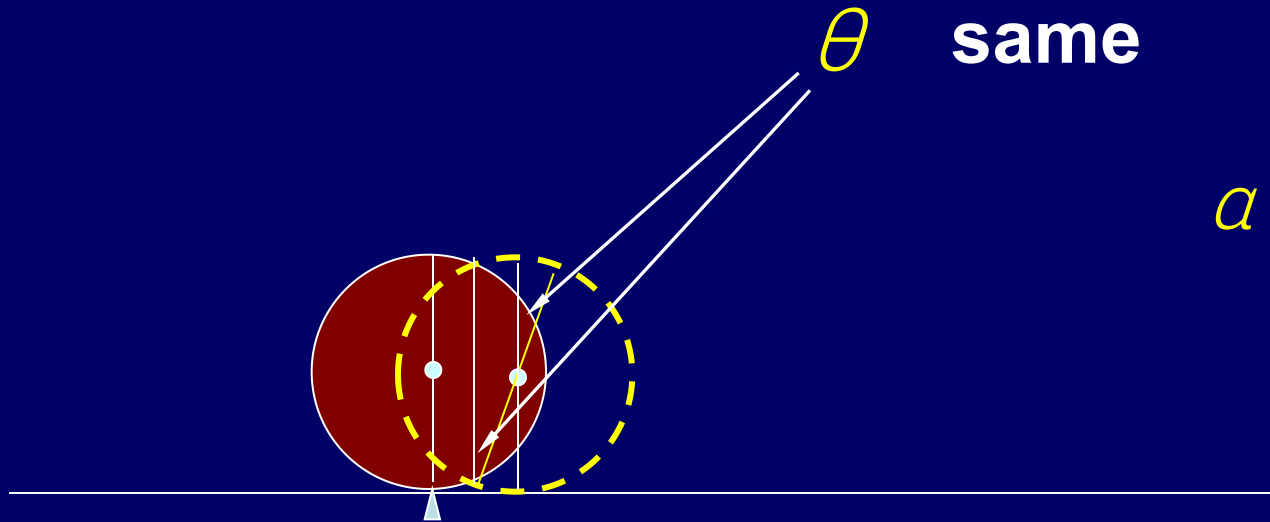
$$mg\sin\theta - f = ma_{cm}$$

$$f = \frac{1}{3}mg \sin\theta$$

Rolling about different axis

$$\omega = \frac{d\theta}{dt}$$

$$a = \frac{d\omega}{dt}$$



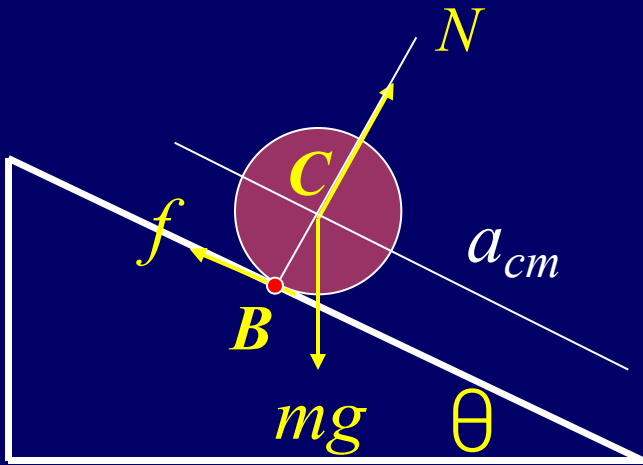
$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$I = \sum m_i r_i^2$$

different

Example

A solid ball of mass m and radius R rolls **without slipping** down an inclined plane with angle θ . Find the a_{cm} and f .



$$mg \sin \theta - f = ma_{cm}$$

$$fR = \frac{2}{5}mR^2\alpha \quad f = \frac{2}{5}mR\alpha$$

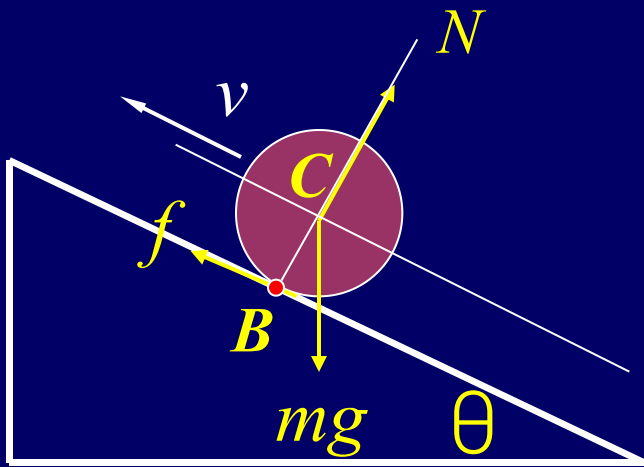
$$a_{cm} = R\alpha \quad f = \frac{2}{5}ma_{cm} = \frac{2}{7}mg \sin \theta \quad a_{cm} = \frac{5}{7}g \sin \theta$$

$$mg \sin \theta = \frac{7}{5}ma_{cm} \quad a_{cm} = \frac{5}{7}g \sin \theta$$

$$mg \sin \theta R = \frac{7}{5}mR^2\alpha$$

$$\alpha = \frac{5}{7R}g \sin \theta$$

Example



A solid cylinder of mass m and radius R rolls with initial speed v . If it rolls **without slipping** up on an inclined plane with angle θ , find the maximum height it can arrive.

$$-Rmg \sin \theta = \left(\frac{3}{2}mR^2\right)\alpha$$

$$\alpha = -\frac{2g}{3R} \sin \theta$$

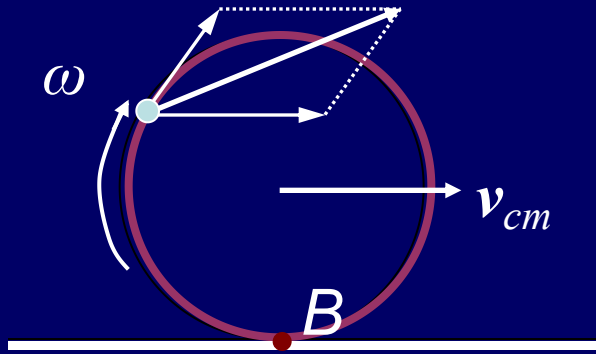
$$a_{cm} = \alpha R = -\frac{2g}{3} \sin \theta$$

$$2a_{cm}L = 0 - v^2$$

$$L = \frac{3v^2}{4g \sin \theta}$$

$$h = L \sin \theta = \frac{3v^2}{4g}$$

Combined rotational and translational motion



Plane parallel motion

$$\Sigma \vec{F}_{ext} = m \vec{a}_{cm}$$

$$\Sigma \tau_{ext} = I \alpha$$

Rolling without slipping

$$\vec{v}_i = \vec{v}_{cm} + \vec{\omega} \times \vec{r}_i'$$

$$v_B = v_{cm} - \omega R = 0$$

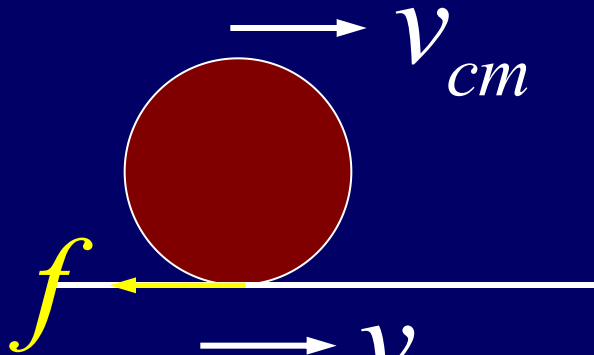
$$v_{cm} = \omega R$$

$$a_{cm} = \alpha R$$

About **B** point

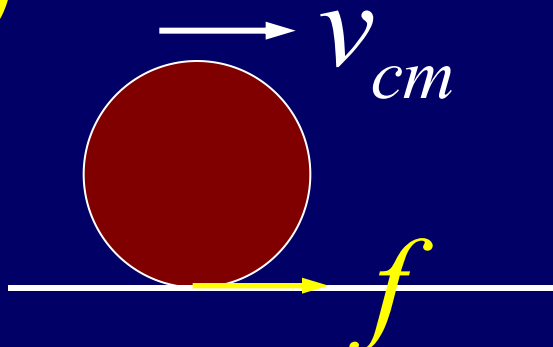
Instantaneous axis

The direction of frictional force in rolling

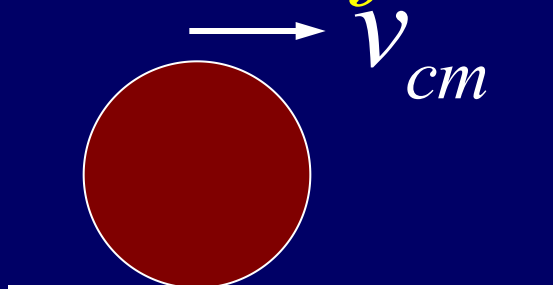


$$\vec{v}_i = \vec{v}_{cm} + \vec{\omega} \times \vec{r}_i'$$

$$v_B = v_{cm} - \omega R$$



$$v_{cm} > \omega R, \quad v_B > 0 \quad \leftarrow f$$



$$v_{cm} < \omega R, \quad v_B < 0 \quad \rightarrow f$$

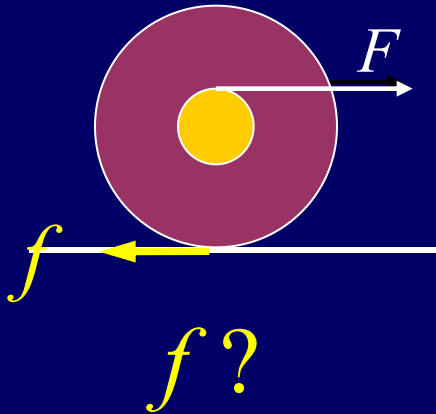
$$f = 0$$

$$v_{cm} = \omega R, \quad v_B = 0 \quad f_k = 0$$

$$f_s = ?$$

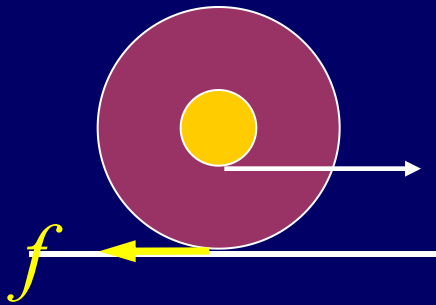
Example

If the force F act on the point $R/4$ from the center, and rotation without slipping



$$F - f = ma$$
$$\frac{F}{4} + f = \frac{1}{2}ma$$

$$f = \frac{F}{6}$$
$$a = \frac{5F}{6m} > 0$$

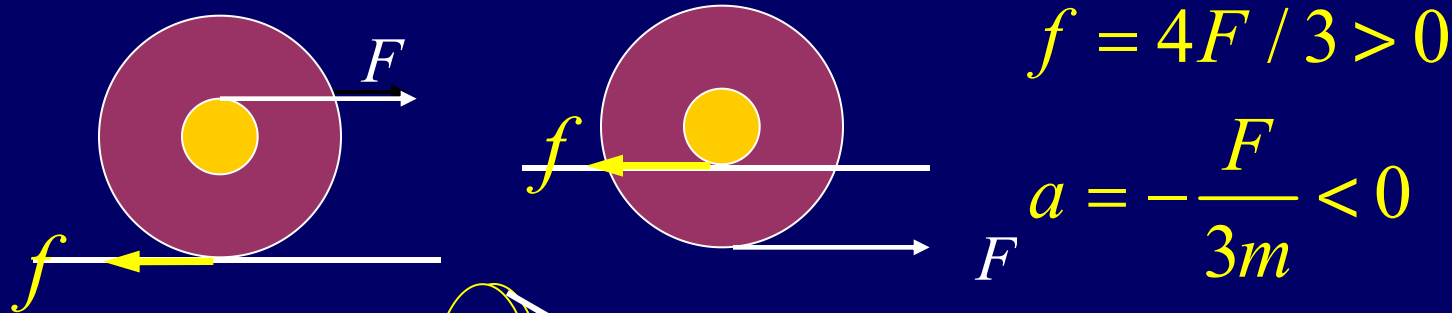


$$F - f = ma$$
$$-\frac{F}{4} + f = \frac{1}{2}ma$$

$$f = \frac{F}{2}$$
$$a = \frac{F}{2m} > 0$$

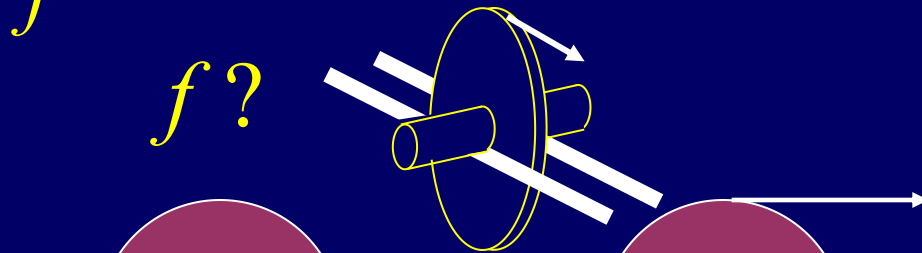
Example

If the force F act on the point $R/4$ from the center, and rotation without slipping



$$f = 4F / 3 > 0$$

$$a = -\frac{F}{3m} < 0$$



$f ?$

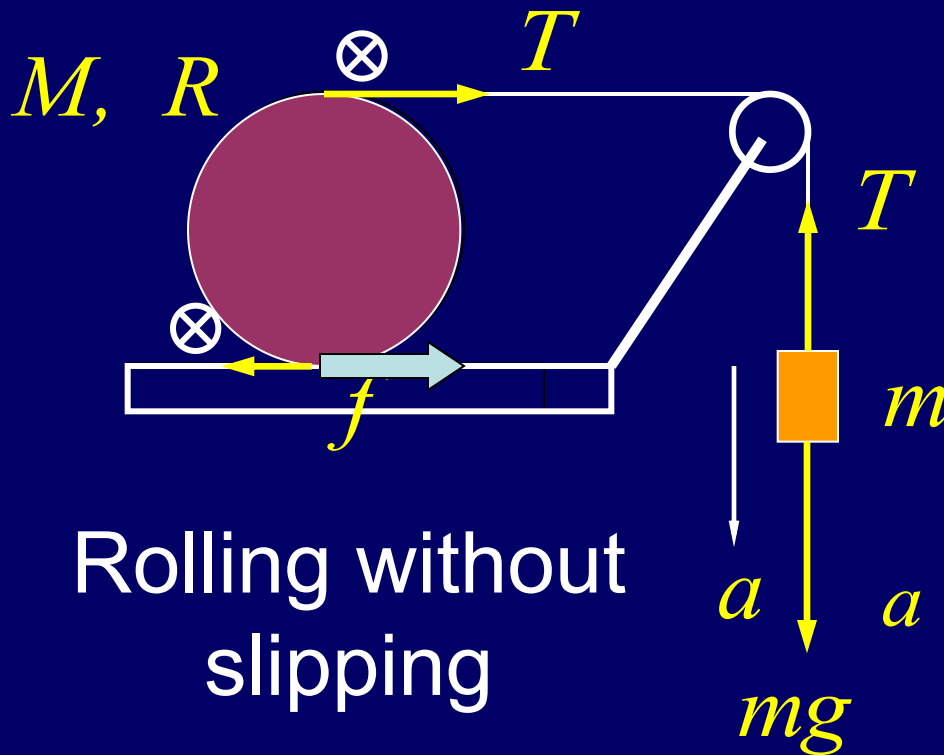
$$F a = \frac{5F}{9m} > 0$$

$$F + \frac{f}{4} = 2ma$$

$$f = \frac{4F}{9} > 0$$



Example



$a = ?$, $T = ?$, $f = ?$

$$mg = ma + \frac{3}{8}Ma$$

$$T - f = Ma_{cm}$$

$$T = \frac{3}{8}Ma = \frac{3mMg}{3M + 8m}$$

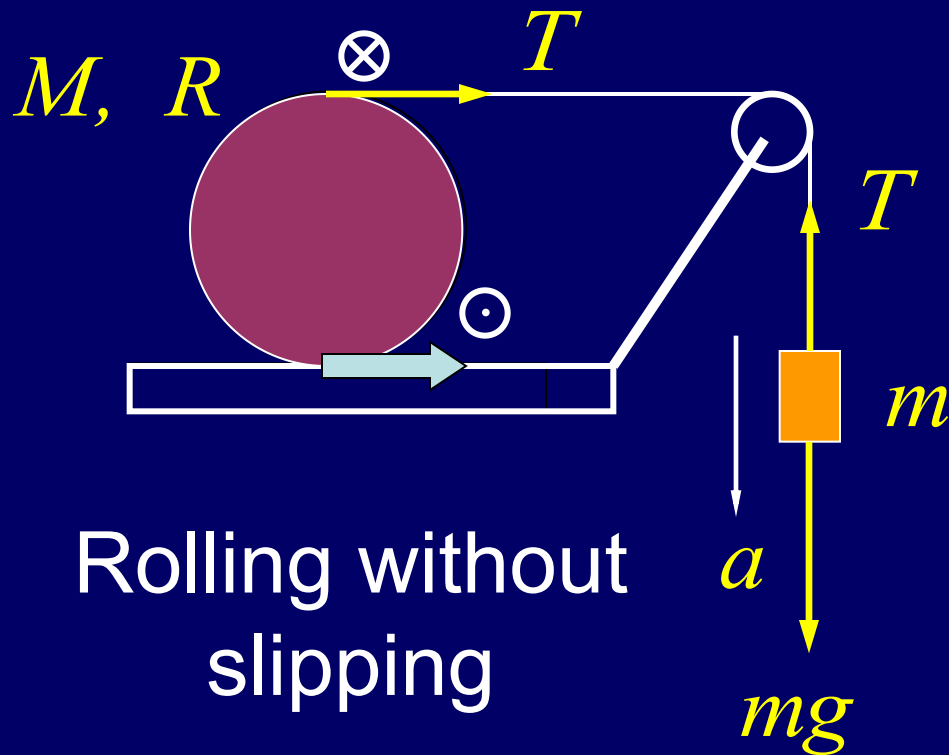
$$aR = a_{cm} \quad a = a_{cm} ?$$

$$a = a_{cm} + aR = 2a_{cm}$$

$$a = \frac{8mg}{3M + 8m} \quad a_{cm} = \frac{4mg}{3M + 8m}$$

$$f = -\frac{Mmg}{3M + 8m}$$

Example



Rolling without
slipping

$a = ?$, $T = ?$, $f = ?$

$$mg - T = ma$$

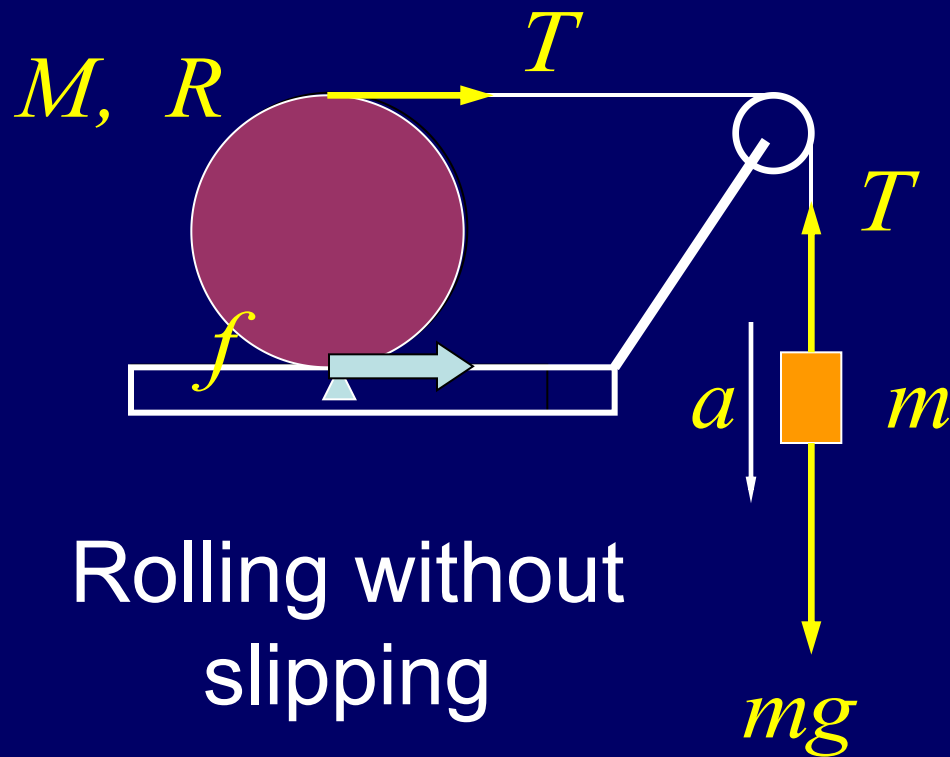
$$T + f = Ma_{cm}$$

$$TR - fR = \frac{1}{2}MR^2 a$$

$$a = \frac{mg}{\frac{3}{8}M + m}$$

$$f = \frac{Mmg}{3M + 8m}$$

Example



$$mg - T = ma$$

$$T = \frac{3}{8}Ma$$

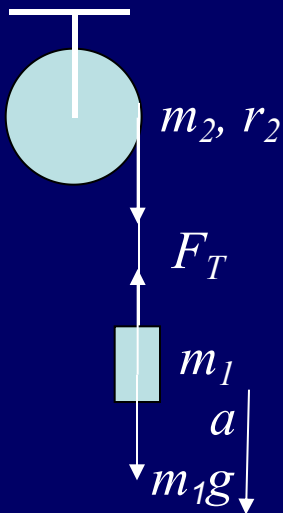
$$2aR = a$$

$$a = \frac{8mg}{3M + 8m}$$

$$T = \frac{3mMg}{3M + 8m}$$

$$a_{cm} = \frac{4mg}{3M + 8m}$$

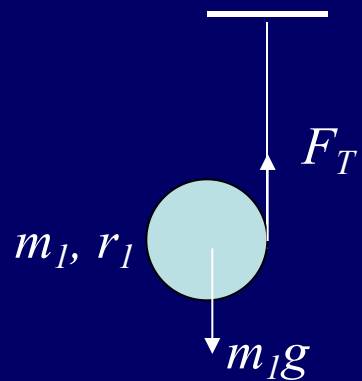
$$\sum F = ma_{cm} \quad T + f = \frac{4Mmg}{3M + 8m} \quad f = \frac{Mmg}{3M + 8m}$$



$$m_1 g - F_T = m_1 a \quad F_T r_2 = \frac{1}{2} m_2 r_2^2 a$$

$$F_T = \frac{1}{2} m_2 r_2 a = \frac{1}{2} m_2 a$$

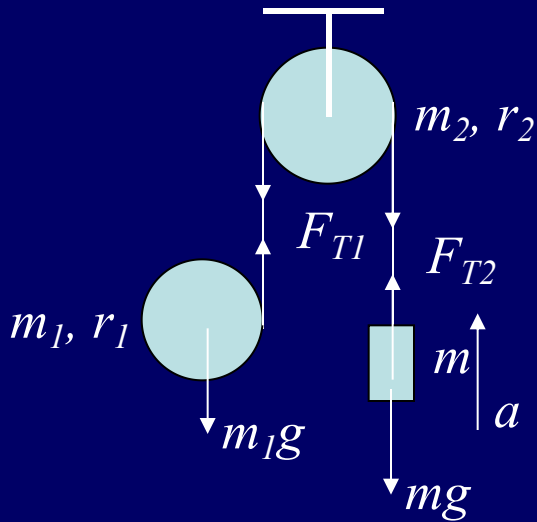
$$m_1 g = m_1 a + \frac{1}{2} m_2 a \quad a = \frac{m_1 g}{m_1 + \frac{1}{2} m_2}$$



$$m_1 g - F_T = m_1 a \quad F_T r_1 = \frac{1}{2} m_1 r_1^2 a$$

$$F_T = \frac{1}{2} m_1 r_1 a = \frac{1}{2} m_1 a$$

$$m_1 g = m_1 a + \frac{1}{2} m_1 a \quad a = \frac{2g}{3}$$



$$F_{T2} - mg = ma \quad (1)$$

$$F_{T1} - F_{T2} = \frac{1}{2}m_2a \quad (2)$$

$$(1) + (2) : F_{T1} - mg = (m + \frac{1}{2}m_2)a \quad (3)$$

$$m_1g - F_{T1} = m_1a_c \quad (4) \quad r_1F_{T1} = \frac{1}{2}m_1r_1^2a_1 \quad F_{T1} = \frac{1}{2}m_1r_1a_1$$

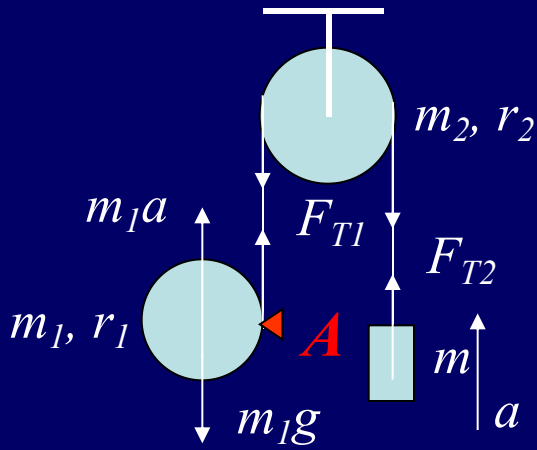
$$(4) \rightarrow a_c = g - \frac{1}{2}r_1a_1 \quad \left. \vphantom{a_c = g - \frac{1}{2}r_1a_1} \right\} a_c = \frac{1}{3}(2g + a)$$

since $a_c = r_1a_1 + a$

$$(3) + (4) : m_1g - mg =$$

$$\frac{2}{3}m_1g + \frac{1}{3}m_1a + (m + \frac{1}{2}m_2)a$$

$$a = \frac{\frac{1}{3}m_1g - mg}{m + \frac{1}{3}m_1 + \frac{1}{2}m_2}$$



$$F_{T2} - mg = ma \quad (1)$$

$$F_{T1} - F_{T2} = \frac{1}{2}m_2a \quad (2)$$

$$(1) + (2) : F_{T1} - mg = \left(m + \frac{1}{2}m_2\right)a \quad (3)$$

*With frame fixed
at A, Inertial
force should be
considered.*

$$m_1g - m_1a - F_{T1} = \frac{2}{3}m_1(g - a) \quad (4)$$

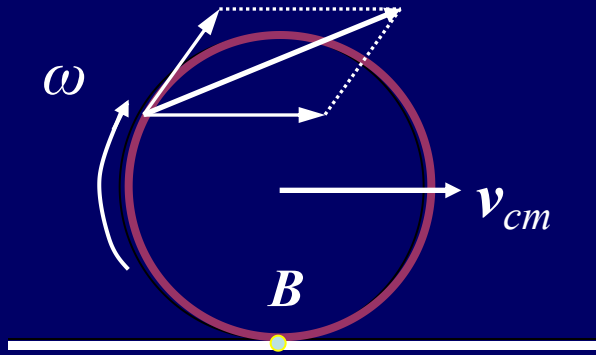
$$r_1m_1g - r_1m_1a = \frac{3}{2}m_1r_1^2a_1 \quad g - a = \frac{3}{2}r_1a_1$$

$$(3) + (4) : m_1g - m_1a - mg = \frac{2}{3}m_1(g - a) + \left(m + \frac{1}{2}m_2\right)a$$

$$\frac{1}{3}m_1g - mg = \left(\frac{1}{3}m_1 + m + \frac{1}{2}m_2\right)a$$

$$a = \frac{\frac{1}{3}m_1g - mg}{m + \frac{1}{3}m_1 + \frac{1}{2}m_2}$$

Combined rotational and translational motion



Plane parallel motion

$$\Sigma \mathbf{F}_{ext} = m \mathbf{a}_{cm}$$

$$\Sigma \tau_{ext} = I \alpha$$

Rolling without slipping

$$\vec{v}_i = \vec{v}_{cm} + \vec{\omega} \times \vec{r}_i'$$

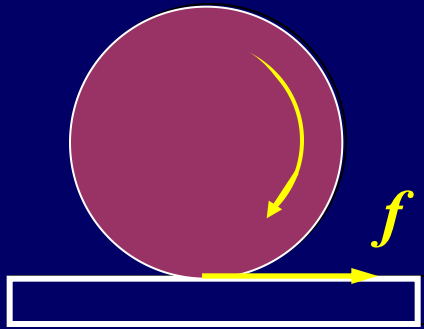
$$v_{cm} = \omega R \quad a_{cm} = \alpha R$$

About **B** point
Instantaneous axis

Rolling with slipping

$$v_{cm} \neq \omega R \quad a_{cm} \neq \alpha R$$

Example



A uniform solid cylinder of radius R and mass M is given an initial (clockwise) angular velocity ω_0 and then lowdown to a uniform horizontal surface.

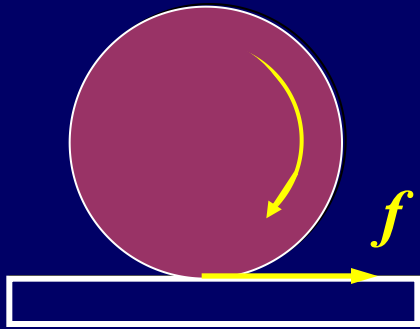
The coefficient of kinetic friction between the surface and the cylinder is μ_k .

Rolling with slipping

Initially the cylinder **slips** as it moves along the surface, but after a time t , pure rolling without slipping begins.

What is the velocity v_{cm} of the center of mass at the time t and what is the value of t ?

Example



A uniform solid cylinder of radius R and mass M is given an initial (clockwise) angular velocity ω_0 and then lowdown to a uniform horizontal surface.

Rolling with slipping The coefficient of kinetic friction between the surface and the cylinder is μ .

$$N - Mg = 0 \quad \left(\frac{1}{2}MR^2\right)a = -fR \quad a = -\frac{2\mu_k g}{R}$$

$$f = Ma_{cm} = \mu_K Mg \quad \omega = \omega_0 + at = \omega_0 - \frac{2\mu_k gt}{R}$$

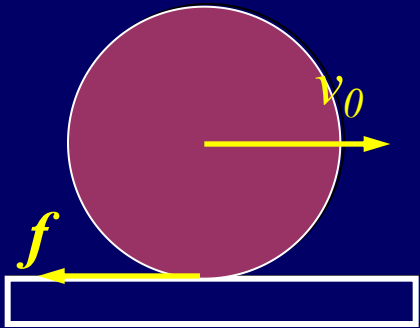
$$a_{cm} = \mu_K g$$

After the time t , the rotation without slipping begins.

$$v_{cm} = \mu_k gt$$

$$\omega = \frac{v_{cm}}{R} \quad t = \frac{R\omega_o}{3\mu_k g} \quad v_{cm} = \frac{R\omega_0}{3}$$

Example



A uniform solid cylinder of radius R and mass M is given an initial velocity v_0 and then lowdown to a uniform horizontal surface.

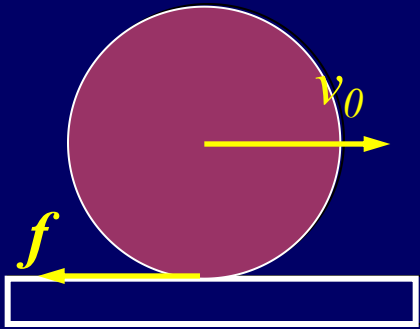
The coefficient of kinetic friction between the surface and the cylinder is μ_k .

Rolling with slipping

Initially the cylinder **slips** as it moves along the surface, but after a time t , pure rolling without slipping begins.

What is the velocity v_{cm} of the center of mass at the time t and what is the value of t ?

Example



A uniform solid cylinder of radius R and mass M is given an initial velocity v_0 and then lowdown to a uniform horizontal surface.

The coefficient of kinetic friction between the surface and the cylinder is μ_k .

$$-\mu_k Mg = -f = Ma \quad a = -\mu_k g$$

$$fR = \frac{1}{2}MR^2 a \quad \mu_k Mg = f = \frac{1}{2}MRa \quad a = \frac{2\mu_k g}{R}$$

$$v = v_0 + at = v_0 - \mu_k gt \quad v = \omega R \quad v_0 - \mu_k gt = 2\mu_k gt$$

$$\omega = at = \frac{2\mu_k gt}{R} \quad t = \frac{v_0}{3\mu_k g} \quad v = \frac{2v_0}{3}$$

CHAP. 9

Exercises

P200: 13, 14, 17, 19, 21
33, 38, 41, 42, 43

problems

P206: 21, 22, 24